DATE RECEIVED (OFFICE USE ONLY)

MAY 03 2007

BOULDER COUNTY
BUILDING DIV

BUILDING PERMIT APPLICATION

Land Use Department - BUILDING SAFETY & INSPECTION SERVICES DIVISION
2045 13th Street / 13th & Spruce Streets / Courthouse Annex
Mailing address: PO Box 471; Boulder, CO 80306
Phone 303-441-3925 / http://www.co.boulder.co.us/lu

(OFFICE USE ONLY)
BP-07- 598

EXPIRED

PROJECT LOCATION

PROJECT STREET ADDRESS	LOT	BLOCK	SUBDIVISION
5134 UTE Highway			

CITY/ZIP CODE	SECTION	TOWNSHIP	RANGE
Lyons 80540	20, 21, 27, 28 & 22	T3N	R70W

OWNERSHIP

☐ CONTACT

OWNER'S NAME	DAYTIME NUMBER
CEMEX INC.	303.823.2100
ADDRESS	EVENING NUMBER
PO Box 1500	
CITY / STATE / ZIP	FAX NUMBER
Houston / TX / 77251	303.823.2199
EMAIL ADDRESS	

CONTRACTOR

☒ CONTACT

CONTRACTOR'S NAME	DAYTIME NUMBER
Cascadia PM, LLC	970-227-6899
ADDRESS	EVENING NUMBER
CITY / STATE / ZIP	FAX NUMBER
	303.648.4229
EMAIL ADDRESS	
Steve.Kidder@cascadiaPM.com	

DESCRIBE THE PROJECT

VALUATION: $ 22,000

Installation & operation of a telecommunications facility to allow three groups of panel antennas and attendant equipment on top of the CEMEX cement plant silo on property located @ 5134 UTE Highway, South of UTE Highway, East of North Foothills Highway.

PROJECT INCLUDES: ☐ PLUMBING ☐ ELECTRICAL ☐ MECHANICAL

TYPE OF PROJECT (CHECK ONE)

(01) ☐ New Structure
(02) ☒ Addition To
(03) ☐ Remodel
(04) ☐ Electrical
(07) ☐ Demolition of Structure
(08) ☐ Change of Use
(09) ☐ Moved-in Structure

TYPE OF STRUCTURE (CHECK ONE)

(01) ☐ One Family Dwelling
(02) ☐ Two Family or Attached Dwelling (including Townhouse)
(03) ☐ Three and Four Family Dwelling
(04) ☐ Hotel or Motel
(06) ☐ Amusement & Recreation Building
(07) ☐ Church & Other Religious Building
(08) ☐ Industrial Building, Manufacturing Plant, Factory
(09) ☐ Service Station & Repair Garage
(10) ☐ Detached Garage
(12) ☐ Office, Bank, and Professional Building
(13) ☐ School
(14) ☐ Store, Customer Service
(15) ☐ Barn, Storage Shed, Outbuilding
(16) ☐ Mobile/Manufactured Home
(22) ☐ Public Works, Utility Building
(23) ☐ Pool, Hot Tub, Fence, Retaining Wall, Pump (Non-structure)
☒ Other: Antennae & radio cabinet install

	EXISTING BUILDING	NEW CONSTRUCTION	EXISTING & NEW TOTAL
Height of Structure	150 FT		
# of Bedrooms	—		
# of Bathrooms	—		
Property Size:	—		

PROPOSED NEW BUILDING OR ADDITION

STRUCTURE SIZE	FINISHED	UNFINISHED
Basement	SQ. FT.	SQ. FT.
1st Floor	SQ. FT.	SQ. FT.
2nd Floor	SQ. FT.	SQ. FT.
3nd Floor	SQ. FT.	SQ. FT.
All other Floors	SQ. FT.	SQ. FT.
Garage	SQ. FT.	SQ. FT.
Deck SQ. FT.	Covered Porch	SQ. FT.

SETBACKS (Distance of Project to Property Lines)

FRONT SETBACK EXISTING	SIDE SETBACK EXISTING	SIDE SETBACK EXISTING	REAR SETBACK EXISTING
FT.	FT.	FT.	FT.
FRONT SETBACK PROPOSED	**SIDE SETBACK PROPOSED**	**SIDE SETBACK PROPOSED**	**REAR SETBACK PROPOSED**
FT.	FT.	FT.	FT.

WATER SERVICE (✓ one)
☐ Individual Well
☐ Community Well
☒ N/A
☐ Public: _____

SEWER SERVICE (✓ one)
☐ Septic Tank
☐ Vault
☒ N/A
☐ Public: _____

ELECTRICAL SERVICE (✓ one)
☐ Xcel Energy
☐ United Power
☐ Estes Park
☒ Poudre Valley Rea
☐ Longmont
☐ N/A

GAS SERVICE (✓ one)
☐ Xcel Energy
☐ Propane
☒ N/A
☐ Other: _____

ACCESS (✓ one)
☒ Existing Driveway
☐ New Driveway
☐ N/A
☐ Other: _____

CERTIFICATION

I certify that the above information is correct and agree to construct this building in accordance with the plot, building plans and specifications submitted herewith, and in strict compliance with all the provisions of the Zoning Ordinace, Building Code, Electrical Code, and Health and Plumbing Regulations of the County of Boulder.

APPLICANT (PLEASE PRINT)	DATE	TITLE
Steven D Kidder	5/3/07	Project Manager

APPLICANT SIGNATURE
Steven D Kidder

BB-07-598

This parcel at 005134 UTE HY
is owned by
CEMEX INC
ATT: TAX DEPT
PO BOX 1500
HOUSTON, TX 77251

Its Legal Description is
21-3N-70, 28-3N-70, 29-3N-70 SE 1/4 SE
1/4 20-3N-70 & PT W 1/2 27-3N-70 & E
500 FT OF LESS .19 ACS M/L TO BOCO &
LESS PTS TO US LESS PT TO HWY DEPT
TOTAL 757.85 ACS SPLITS 128969 & 142930
TO HWY DEPT 005134 UTE HY LONGMONT□

Its GIS Land Area is 37738557 square feet, 866.358 acres.

It is Parcel Number 120328000001 and this owner has Account Number R0053049.

These owners are taxed in the following:
BOULDER COUNTY GENERAL OPERATING
ST VRAIN RE1J GENERAL OPERATING
NORTHERN COLO WATER CONTRACT OBLIG
ST VRAIN LEFT HAND WATER GENERAL OPER
HYGIENE FIRE DIST GENERAL OPERATING
LONGMONT CONSERVATION DIST GEN OPER
RTD GENERAL OPERATING

Recent deeds for this parcel:
SW at Reception Number 613148 from to for $42000000 on 4/5/1984
QD at Reception Number 1284471 from to for $0 on 4/21/1993
QD at Reception Number 1284471 from to for $0 on 4/21/1993
SW at Reception Number 1432787 from to for $818000 on 6/1/1994
SW at Reception Number 1432787 from to for $818000 on 6/1/1994
SW at Reception Number 1477357 from to for $0 on 11/7/1994
SW at Reception Number 1477357 from to for $0 on 11/7/1994
SW at Reception Number 1756839 from to for $0 on 12/19/1997

NOTE: Because of small inconsistencies in the locations of lines in the map layers, GIS searches may return data from adjacent polygons even though no overlap is visible on the screen. Display the layers in question to resolve any uncertainty.

PLSS Section, Township, Range on or adjacent to this Parcel:
T3NR70WS20

T3NR70WS21
T3NR70WS29
T3NR70WS28
T3NR70WS27

Zoning Districts on or adjacent to this Parcel:
GI
GI with Floodplain Overlay
A with Floodplain Overlay
A with Floodplain Overlay
A
A

Open Space ownerships on or adjacent to this Parcel:
CE BRALY SE SUB
COS WESTERN MOBILE
COS MONTGOMERY (DONALD)
COS WESTERN MOBILE
COS WESTERN MOBILE
COS WESTERN MOBILE

County Plats on or adjacent to this Parcel:

Permits for this Parcel:
BP-00-872 ELEC FOR WATER PUMP MAIN PLANT

BP-00-1207 CELLULAR ANTENNA SITES

BP-01-494 SIGN REPLACEMENT

BP-01-2179 INSTALL PCS EQUIP & ANTENNAS ON CEMENT PLANT - FINAL INSPECTION COMPLETED ON 2/11/02 BY J.H.

BP-04-1389 TEMPORARY OFFICE TRAILER

BP-04-1670 STOREROOM MODIFICATIONS

BP-05-129 REPLACE EXISTING CONDITIONING TOWER

BP-06-692 INSTALL TWO 100GAL WATER HEATERS IN THE OFFICE BLDG

BP-65-8040 EARTHWORK & FOUNDATIONS FOR CEMENT PLANT

BP-66-9358 NEW RES

BP-67-10137 SUPERSTRUCTURE FOR CEMENT PLANT

BP-70-13156 SIGN FOR CEMENT PLANT

BP-73-18153 COAL STORAGE SILO

BP-75-19820 SHOP

BP-78-22476 REPAIR OVERHEAD DOOR

BP-78-23408 ELEC FOR NEW PUMP

BP-79-324 ENLARGE STORAGE BLDG

BP-79-463 COMBUSTION CHANGE/KILN

BP-79-513 5 CONSTRUCTION MOHOS

BP-80-619 ELEC FOR MOBILE HOME

BP-81-71 elec for traffic light

BP-84-182 FOUNDATION FOR CLINKER COOLER

BP-90-230 REPAIR GAS LEAK/NEW WATERLINE

BP-94-2403 DEMO-DWELLING,BARN,GAR,SHED,SILO

BP-94-2566 OFFICE TRAILER

BP-97-1785 TELECOM SITE/WALL MOUNTED

Dockets for this Parcel:
NWE-05-012 CEMEX INC.

PAC-03-032 CEMEX INC.

SE-96-18 WESTERN MOBILE BOULDER, INC.

SPR-97-160 SOUTHDOWN INC / US WEST NEWVECTOR G

SU-01-08 QWEST WIRELESS (Cemex Cement Plant)

SU-06-016 CINGULAR WIRELESS (Cemex Inc.) Tele

SU-88-21 LYONS RUBBER RECYCLING CENTER

SU-96-18 WESTERN MOBILE BOULDER, INC.

ZON-07-028 CEMEX INC

Assessor land and structure listing for this parcel: XFOB = Extra Feature or Building

ACCOUNTNO	MODEL	OCCURANCE	ATRIBTYPE	ATRIBDESC
R0053049	COMML	1.0	ABSTRACT_CODE	MANUFACTURING/PRO IMPRO
R0053049	COMML	1.0	CONST	TEES/TILT
R0053049	COMML	1.0	DEPR01	1969
R0053049	COMML	1.0	DESIGN	MASONRY
R0053049	COMML	1.0	HEIGHT	42 - HEIGHT
R0053049	COMML	1.0	QUALITY	AVERAGE
R0053049	COMML	1.0	STORIES	1 - STORIES
R0053049	COMML	1.0	YEAR	1960 - YEAR
R0053049	LAND_COMML	1.0	ABSTRACT_CODE	REFINING/MILLING-LA
R0053049	LAND_COMML	100.0	ABSTRACT_CODE	MANUFCTNG/PROCESN
R0053049	LAND_COMML	101.0	ABSTRACT_CODE	REFINING/MILLING-LA
R0053049	LAND_MT	3.0	ABSTRACT_CODE	SINGLE FAM.RES.-LANI
R0053049	RESMT	3.0	ABSTRACT_CODE	SINGLE FAMILY RESIDI IMPR
R0053049	RESMT	3.0	BATH_F	1 - BATH_FULL
R0053049	RESMT	3.0	BEDROOMS	BEDROOMS
R0053049	RESMT	3.0	CONST	FRAME
R0053049	RESMT	3.0	DEPR01	YEAR
R0053049	RESMT	3.0	DESIGN	1 STORY - RANCH
R0053049	RESMT	3.0	QUALITY	AVG PLUS
R0053049	RESMT	3.0	ROOMS	ROOMS
R0053049	RESMT	3.0	YEAR	1960 - YEAR
R0053049	XFOB	301.0	ABSTRACT_CODE	SINGLE FAM RES IMPROVEMENTS
R0053049	XFOB	301.0	DEPR01	YEAR
R0053049	XFOB	301.0	XFOB_CODE	BARN-D,LOW 15 YRS
R0053049	XFOB	301.0	YEAR	1960 - YEAR

Assessor land and structure areas or counts for this parcel: Correlate with land and structures by Model+Occurance for each Account.

ACCOUNTNO	MODEL	OCCURANCE	SUB_CODE	AREA_ACT
R0053049	COMML	1.0	4940	181928.0
R0053049	LAND_COMML	1.0	SQFT	1829520.0
R0053049	LAND_COMML	100.0	SQFT	1873080.0
R0053049	LAND_COMML	101.0	ACRES	232.0
R0053049	LAND_MT	3.0	ACRES	440.85
R0053049	RESMT	3.0	AREA_FIRST	1508.0
R0053049	RESMT	3.0	B_SUB_UNF	1508.0
R0053049	RESMT	3.0	POR_PORCH	260.0
R0053049	XFOB	301.0	XFOB_UNITS	840.0

Building Permit Routing Form

BP-07- 598

Permit tech SL

Owner Name Cemex

Project Description install 3 groups of panel antennas

Fees:

Application Deposit	$ 25.00 pd ✓	Subtotal	$ 654.88
Building Division Fees	$ 654.88	Less deposit	- $ 25.00
Investigation Fee	$	BALANCE DUE:	$ 629.88 Balance due
Access Fee	$		
Escrow	$ Escrow $		
Floodplain Fee	$		

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Permit tech comments:

Zoning Approval

SPR req'd____ zoning affidavit____ height survey req'd____ setback survey req'd____
Comments:
SU-06-016

approvedby/date
5-3-07 RAH/DSC

Historic Approval

Comments:

approvedby/date

Site Plan Review

Site Plan Review #_____ notified owner/date:_____ notified contractor/date:_____
Comments:

approvedby/date

Transportation Approval

access permit #_____ road maint waiver req'd___ inspection req'd prior to CO___
Comments:

approvedby/date

Floodplain

floodplain permit #_____ Comments:

approvedby/date

Plan Check

Soils Report Info: Caiss Spd Ftgs Per.Drain Voids_____ Float.Slabs_____
Comments:

JPH approvedby/date

PUT ON JON'S DESK 5.9.07 MBH 5.16.07

Chief Building Official

Comments:

JPH 5/16/07 approvedby/date

SPR zoning transportation

well/water septic/sanitation access permit road maint waiver

notification: called (whom?_____) sent card mailed permit

BUILDING PERMIT APPLICATION REVIEW FORM

FILE NUMBER: 07-598

PROJECT LOCATION

PROJECT ADDRESS	UNIT	CITY	ZIP CODE
5134 Ute Hwy		~~LY~~	80540

LOT	BLOCK	SUBDIVISION NAME	SECTION	TOWNSHIP	RANGE
			28	3N	70

OWNER'S NAME: Conrey Inc

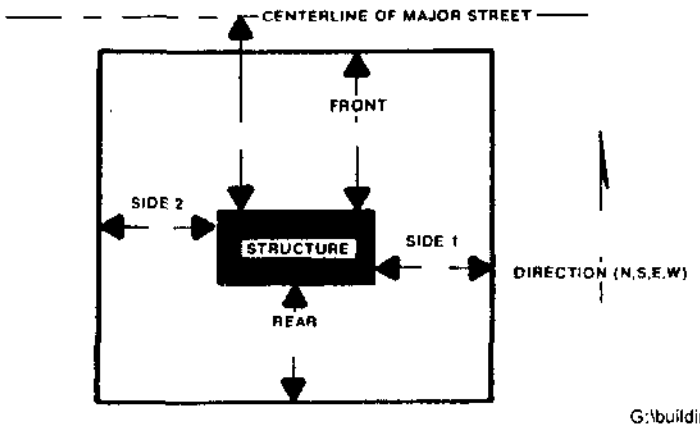
OCCUPANCY GROUP	TYPE OF CONSTRUCTION	VALUATION
U	VB	$ 22,000

STRUCTURE DATA	EXISTING	TO BE CONSTRUCTED
Height of Structure	FT	FT
Number of Stories		0
Number of Dwellings		↓
Number of Bedrooms		↓
Number of Bathrooms		↓

Water	
Electric	
Sewer	
Gas	
Access	

STRUCTURE SIZE (For new construction & additions only)	FINISHED	UNFINISHED
Basement	SQ FT	SQ FT
1st Floor		
2nd Floor		
All other Floors		
Deck		
Covered Roof		
Garage		
Totals		
Snowload		
Windload		

Tax I.D.#	
Parcel #	1263 28 0 00 001
Deed #	
Tax Area	
Service Area	
School District	
Fire District	
Property Size	ACRE(S)
Zoning (1)	ACRE(S)
Zoning (2)	ACRE(S)



BP-07-598

RESOLUTION 2006-158

A RESOLUTION CONDITIONALLY APPROVING BOULDER COUNTY LAND USE DOCKET #SU-06-016 ("CINGULAR WIRELESS (CEMEX, INC.) TELECOMMUNICATIONS FACILITY SU/SSDP"): A SPECIAL USE REVIEW WITH ASSOCIATED SITE SPECIFIC DEVELOPMENT PLAN REQUEST FOR THE INSTALLATION AND OPERATION OF A TELECOMMUNICATIONS FACILITY TO ALLOW THREE GROUPS OF PANEL ANTENNAE AND ATTENDANT EQUIPMENT ON TOP OF THE CEMEX CEMENT PLANT SILO, ON PROPERTY LOCATED AT 5134 UTE HIGHWAY, SOUTH OF UTE HIGHWAY, EAST OF NORTH FOOTHILLS HIGHHWAY, IN SECTIONS 20, 21, 27, 28, AND 29, T3N, R70W

WHEREAS, Cingular Wireless ("Applicant") has requested approval for a special use permit, with associated site specific development plan, under Article 4-600 of the Boulder County Land Use Code ("the Land Use Code"), to install a telecommunications facility on top of the CEMEX cement plant which is located on the property as described in the caption to this Resolution, above ("the Subject Property"), in the General Industrial Zoning District in unincorporated Boulder County; and

WHEREAS, previously, the County has reviewed and approved the following telecommunications facilities in this same location: Docket #SU-01-08: Qwest Wireless; Docket #SU-96-30, Southdown, Inc. Roof Top Antenna; and Docket #SPR-97-160; and

WHEREAS, the Applicant now proposes to place three sets of four panel antennae at the top of the existing silo; and

WHEREAS, one set of antennae will face west and be placed flush to the side of the silo at a height of about 145 feet (to the centerline of the antennae); another set of antennae will be placed on the top of the existing service elevator, which extends above the existing silo, and will face east, being placed at a height of about 155 feet (measured to the centerline of the antennae); and a third antennae set will face south, being at a height of 167 feet (measured to the centerline of the antennae) and placed on a frame extending above the existing dust collector penthouse, which itself extends 10 feet above the roof of the silo; and

WHEREAS, all of the antennae will be painted the same color as the existing silo.

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BOULDER COUNTY
BUILDING DIV.

07-598
5134 Ute
Hwy

WHEREAS, the above-described request was processed and reviewed as Boulder County Land Use Docket #SU-06-016 ("the Docket"), all as further described in the Boulder County Land Use Department Planning Staff's Memorandum and written recommendation to the Boulder County Board of County Commissioners ("the Board") dated December 7, 2006, with its attachments ("the Staff Recommendation"); and

WHEREAS, on November 15, 2006, the Boulder County Planning Commission ("the Planning Commission") held a duly-noticed public hearing on the Docket, and recommended conditional approval of the Docket to the Board; and

WHEREAS, on December 7, 2006, the Board held a duly-noticed public hearing on the Docket ("the Public Hearing"), at which time the Board considered the Staff Recommendation and the recommendation of the Planning Commission, and also considered documents and testimony presented by the County Land Use Department Staff and a representative of the Applicant, with no members of the public being present to speak; and

WHEREAS, based on the Public Hearing, the Board finds that the Docket meets the criteria for special use approval set forth in Article 4-600 of the Land Use Code, and can be approved, subject to the conditions stated below, and finds further that the Docket, with the submission of the standard development agreement, meets the criteria in the Land Use Code for a site-specific development plan, subject to the conditions stated below.

NOW, THEREFORE, BE IT RESOLVED that the Docket is hereby approved, on the basis set forth in this Resolution, above, and subject to the following conditions:

1) The antennae, and all support structures shall be mounted to the silo as noted in the application documents for the Docket.

2) The color of the antennae shall match the existing silo. Before the issuance of a building permit, the Applicant shall provide a color sample for review and approval by the Land Use Department.

3) The heights of the proposed antennae shall not exceed that which is depicted in the application materials for the Docket.

4) The Applicant shall submit a Development Agreement to be reviewed and approved by the County Attorney and Land Use staff before its recordation. The vested right granted by this site specific development plan approval shall run for a period of three years, beginning on the date of adoption of this Resolution as set forth below. The vested right shall expire immediately upon the running of this three-year period, unless the Board in its discretion approves an extension pursuant to the applicable provisions of the Land Use Code.

5) There shall be a review hearing every seven years from the date of approval of this Special Use, and at seven-year intervals thereafter, to consider evolving telecommunication technologies and alternatives. During the hearing, if it is determined that the telecommunication facility approved herein is no longer needed, the Applicant/owner of such telecommunication facility shall remove the facilities at their own expense.

6) The Applicant shall agree to allow collocation of other telecommunication providers, if reasonable accommodations can be reached and subject to approval according to the applicable County land use review process.

7) The Applicant shall be subject to the terms, conditions, and commitments of record for the Docket.

A motion to approve the Docket, as stated above, was made by Commissioner Toor, seconded by Commissioner Pearlman, and passed by a 2-0 vote, with Commissioner Mayer being excused.

ADOPTED this ~~21st~~ 22nd day of December, 2006, nunc pro tunc the 7th day of December, 2006.

BOARD OF COUNTY COMMISSIONERS OF BOULDER COUNTY:

Ben Pearlman, Chair

Thomas A. Mayer, Vice Chair

Will Toor, Commissioner

ATTEST:

Clerk to the Board

Land Use

Courthouse Annex - 2045 13th Street - Boulder, Colorado 80302 - (303) 441-3930 - Fax 303-441-4856
Mailing Address: Post Office Box 471 - Boulder, Colorado 80306 **www.bouldercounty.org**

Cindy Domenico *County Commissioner* **Ben Pearlman** *County Commissioner* **Will Toor** *County Commissioner*

January 5, 2011

Name: Cemex Inc

Address: 5134 Ute Hw

Longmont, CO 80503

Re: Building Permit BP-07-0598 for INSTALL 3 GROUPS OF PANEL ANTENNAS at 5134 Ute Hw,

Cascadia Pm Llc

A routine review of our building permit records shows that the work covered by the referenced building permit has not been inspected for at least six months and has not received final inspection approval. Boulder County Building Code Section 109 requires that you obtain final inspection approval before the space covered by the permit is used or occupied.

Boulder County Building Code section 105 provides that a building permit shall expire by limitation and become invalid if the work authorized by the permit is suspended or abandoned for a period of 180 days (6 months) or more. For this reason we want to notify you that your permit may be close to expiring or may have expired. It is certainly in your best interest to obtain the required inspection approvals and we would like to assist you in doing so.

We are here to serve as a resource to you. It is important to have a record of inspection approvals and authorization to usé and occupy the space covered by the permit. This will enable you to obtain financing, insurance, and to conduct future real estate transactions.

After reading the above information please contact our office at 303-441-3925 to schedule an inspection or provide a written status report. This report should indicate what work has taken place in the past six months, what work is planned for the next six months, and an indication that the space covered by the permit is not being used or occupied.

We feel that it is your responsibility to resolve this matter. If we do not receive a response within 30 days your building permit file will be closed.

Thank You

Building Safety and Inspection Services Division

CC: Cascadia Pm Llc

Land Use

Courthouse Annex - 2045 13th Street - Boulder, Colorado 80302 - (303) 441-3930 - Fax 303-441-4856
Mailing Address: Post Office Box 471 - Boulder, Colorado 80306 **www.bouldercounty.org**

Cindy Domenico *County Commissioner* **Ben Pearlman** *County Commissioner* **Will Toor** *County Commissioner*

March 7, 2011

Name: Cemex Inc

Address: 5134 Ute Hw

Longmont, CO 80503

Re: Building Permit BP-07-0598 for INSTALL 3 GROUPS OF PANEL ANTENNAS at 5134 Ute Hw,

Cascadia Pm Llc

On January 5, 2011 you were notified by the Boulder County Building Division that the referenced building permit was in danger of becoming expired. Since you have failed to respond to our request to notify us of the status of your project, we must assume that your permit has expired. Boulder County Building Code section 105.5 states that a building permit shall expire by limitation and become invalid if work authorized by the permit is not started or is suspended or abandoned for a period of 180 days (6 months) after the permit is issued. Before such work may be recommenced, a new permit must first be obtained and the required fees paid.

Boulder County Building Code section 109 provides that the work described on the permit must be inspected and receive final inspection approval before the building may be used or occupied. The Boulder County Building Division and the Boulder County Attorney's Office are very concerned about the possibility of a building being occupied without having undergone the required inspections and without having received the required final inspection approval. In the event that this type of illegal occupancy is taking place, there is a strong possibility that action will be taken to prevent it or to cause it to cease.

The Boulder County Building Division feels that it has fulfilled its obligation to notify you of possible violations to the County code. Correcting these violations is your responsibility. You will also be responsible for and must be willing to accept any repercussions of action, which the County may take to force the correction of these violations. We encourage you to contact this office regarding the status of this building permit. If you fail to contact us within the 30 days, your building permit file will become expired and there is a strong possibility of action by the Boulder County Attorney to force you to correct the violations to the county code.

Thank You

Building Safety and Inspection Services Division

CC: Cascadia Pm Llc

Boulder County Land Use Department
Building Safety & Inspection Services Division
Courthouse Annex Building • 2045 13th Street • P.O. Box 471
Boulder, Colorado 80306
t: (303) 441-3925 • f: (303) 441-4856
e: building_official@co.boulder.co.us • www.co.boulder.co.us/lu/
Office Hours: Monday - Friday 8:00 AM to 4:30 PM

Correction Notice

Owner/Project Name	Building Permit Number
CEMEX	BP-07-598
Project Address	**Inspector**
5734 UTE HWY	JON HARMAN
	Date
	12-21-07

Important Note: The corrections listed below must be completed, reinspected and approved prior to proceeding any further or covering any work. Please phone (303) 441-3925 for a reinspection when all corrections have been completed. • Thank You

☐ **If this box is checked, a $69.00 reinspection fee must be paid** at the building safety division office before any additional inspections will be scheduled. Please make sure that work is ready for inspection prior to scheduling inspections. • Thank You

Inspection Requested: METER & RFE

Approved (Inspector)	Date	Item #	Corrections
		X	THE GROUNDING ELECTRODE CONDUCTOR SHALL ATTACH TO THE GROUNDING ELECTRODE WITHOUT SPLICE.
		2.	INSTALL A WIRE SUPPORT AT THE TOP LB,
		X	THE 26½' CLEARANCE NEEDS TO BE CLEARLY VERIFIED WITH SOME MEANS OTHER THAN DANGLING A TAPE MEASURE.
		4.	PLACARDS ARE REQUIRED BEFORE FINAL FOR EACH FEEDER SUPPLYING THE BUILDING — SEE 225.37. PLEASE READ THIS SECTION.
		X	BUSHINGS ARE REQUIRED AT GRC ENDS.
		X	LIGHTS ARE REQ TO BE DOWN-LIT REMOVED
			AT FINAL ELEC INSPECTION

JH 12/21/07

correction notice.cdr • 01.16.07

White - Project Copy • Canary - Inspectors Copy

at&t WIRELESS

REVIEWED
MAY 16 2007
BOULDER COUNTY
BUILDING DIVISION

SITE NAME: LYONS
SEARCH RING NO.: 1469

PROJECT DATA

LATITUDE: 40° 12' 12.6"
LONGITUDE: 105° 14' 8.16"

ROOF TOP ELEVATION: 5360' AMSL

JURISDICTION: BOULDER COUNTY, COLORADO

OCCUPANCY GROUP: U-2
CONSTRUCTION TYPE: V-B

DESIGN WIND LOAD: 120 MPH, EXPOSURE C, 3-SEC GUST

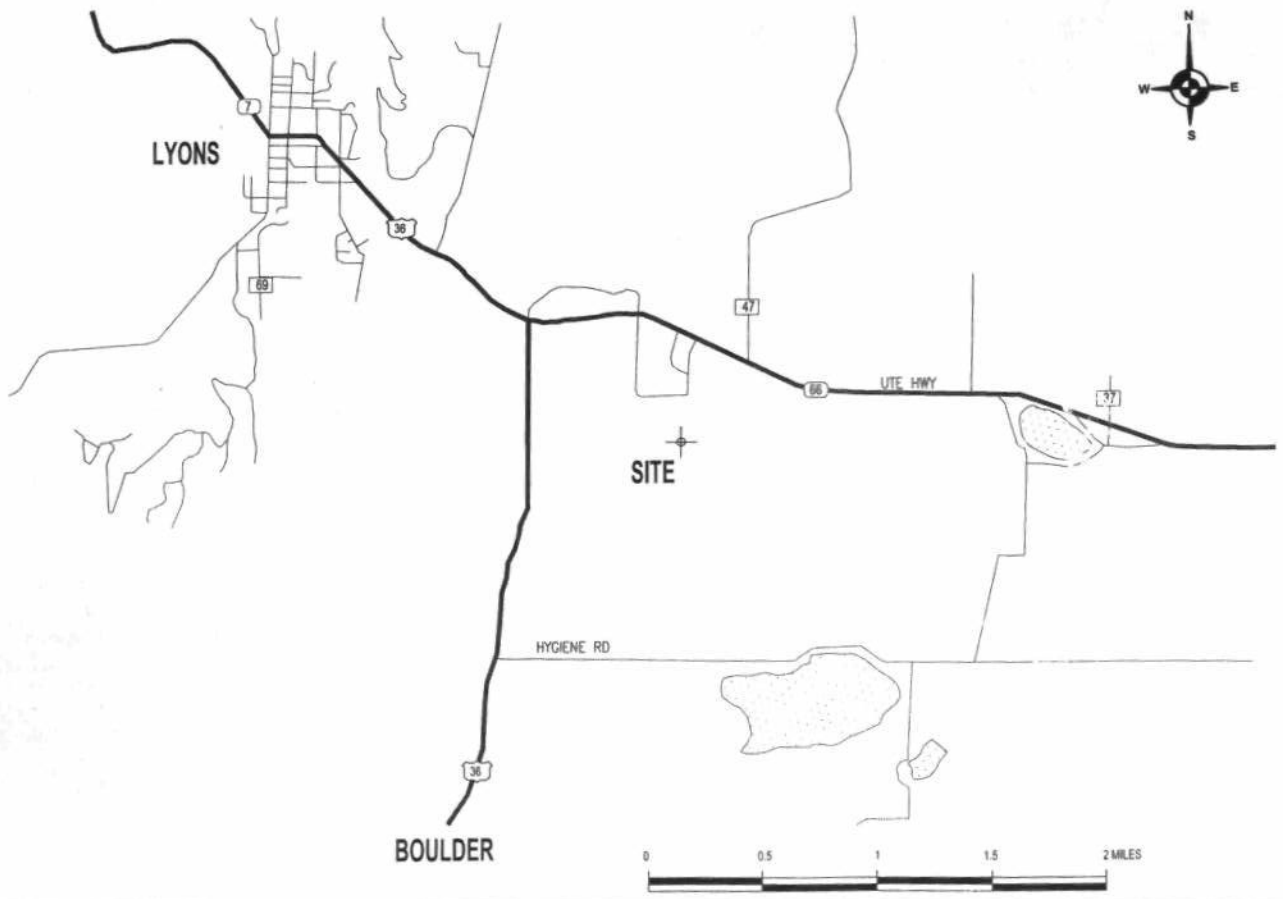
GOVERNING CODES

IBC-2003, INTERNATIONAL BUILDING CODE

NEC-2002, NATIONAL ELECTRICAL CODE

A.D.A. COMPLIANCE
THIS IS AN UNMANNED FACILITY AND NOT FOR HUMAN HABITATION.

VICINITY MAP



PROJECT CONTACTS

APPLICANT
AT&T WIRELESS
188 INVERNESS DRIVE WEST
ENGLEWOOD, CO 80112

PROPERTY OWNER
CEMEX
5145 UTE HWY
LYONS, CO
CONTACT: RON SMITH, 303.823.2199

PROJECT MANAGER
STEVE KELLY
CASCADIA PM
5950 SOUTH WILLOW DRIVE, SUITE 304
GREENWOOD VILLAGE, CO 80111
PH: 303.694.7600

CONSULTING ENGINEER
CHRIS PORCELLO, PE
PEI
6810-B NORTH BROADWAY
DENVER, CO 80221
PH: 303.412.8900

PROJECT SUMMARY

THIS PROJECT INCLUDES THE FOLLOWING SCOPE OF WORK:

1. PLACEMENT OF AT&T PROPRIETARY RADIO EQUIPMENT CABINETS ON THE ROOF OF THE 150-FT TALL SILO CLUSTER. THE SILOS ARE CONSTRUCTED OF REINFORCED CONCRETE, INCLUDING THE ROOF STRUCTURE. THE EQUIPMENT SHALL BE PLACED ONTO A PRE-MANUFACTURED GRAVITY MOUNTED PLATFORM ON THE SILO ROOF.

2. INSTALLATION OF ANTENNAS AT THREE LOCATIONS, AS SHOWN ON THESE DRAWINGS WITH ASSOCIATED CONDUIT AND COAXIAL CABLING SURFACE MOUNTED TO EXISTING HANDRAILS AND VARIOUS STRUCTURES ON THE ROOF OF THE SILOS.

3. ELECTRICAL SERVICE SHALL BE SUPPLIED FROM THE SERVICE POLE MOUNTED TRANSFORMER LOCATED ON THE GROUND LEVEL AT THE RAILROAD TRACKS TO THE EAST SIDE OF THE SILOS.

4. INSTALLATION OF AN ANTENNA FRAMES TO SUPPORT (6) NEW AND (6) FUTURE ANTENNAS AT THE SPECIFIED ELEVATION INCLUDING COAXIAL CABLES INSTALLED FROM THE ROOF MOUNTED EQUIPMENT TO THE ANTENNAS.

DRAWING INDEX

SHEET	DESCRIPTION
T-1	TITLE SHEET
C-1	SITE (ROOF) PLAN
C-2	ENLARGED ROOF PLAN
C-3	ELEVATIONS
C-4	DETAILS
E-1	ELECTRICAL PLAN
E-2	GROUNDING PLAN AND ONE-LINE DIAGRAM
E-3	ELECTRICAL DETAILS
E-4	ELECTRICAL DETAILS
GN-1	GENERAL CONSTRUCTION NOTES
GN-2	GENERAL CONSTRUCTION NOTES

APPROVAL LIST

TITLE	SIGNATURE	DATE
SITE OWNER		
RF ENGINEER		
AT&T WIRELESS		
CASCADIA PM		

DIRECTIONS TO SITE

FROM DENVER, TAKE I-25 NORTH TO HWY 36, FOLLOW 36 WEST THROUGH BOULDER. TURN RIGHT ONTO UTE HWY (RT 66) AND GO EAST APPROXIMATELY 0.5 MILES. SITE IS ON THE SOUTH SIDE OF UTE HWY.

at&t

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6810-B NORTH BROADWAY
DENVER, CO 80221
Phone 303.412.8900 Fax 303.412.8999
Consulting Engineers

COLORADO REGISTERED
CHRISTOPHER PORCELLO
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5-1-07
PROFESSIONAL ENGINEER

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REV	DATE	DESCRIPTION	DRAWN BY	CHECKED BY	APP'D BY
1	4/30/07	CD PERMIT APPLICATION			

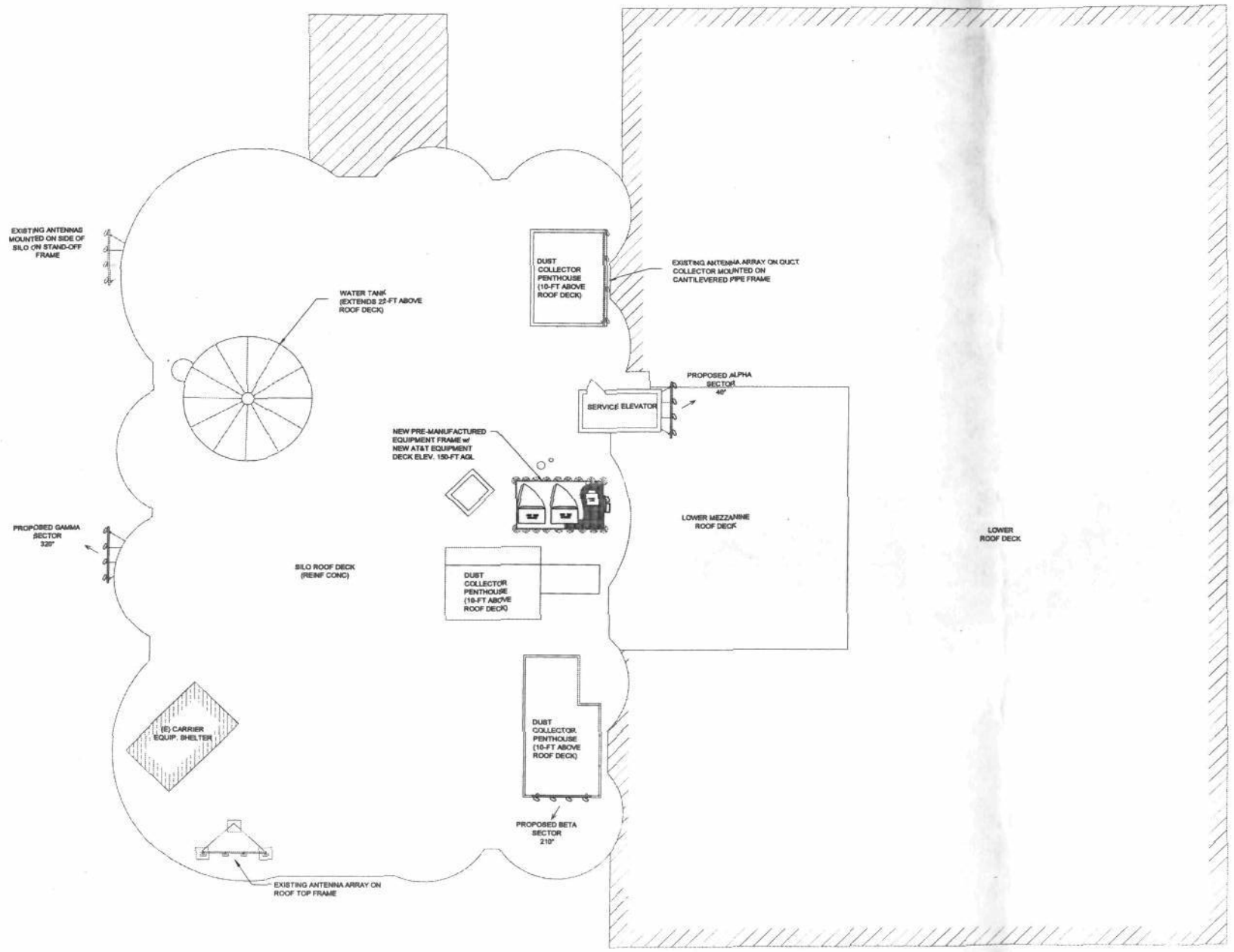
SITE NO.
1469

SITE NAME
LYONS

SITE ADDRESS
5145 UTE HWY
LYONS, CO

SHEET TITLE
TITLE SHEET

SHEET NO.
T-1
SHEET 1 OF 11



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BUILDING DIVISION

N
W E
S

SCALE
3/32" = 1'-0"

SITE PLAN

at&t

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DENVER, CO 80221
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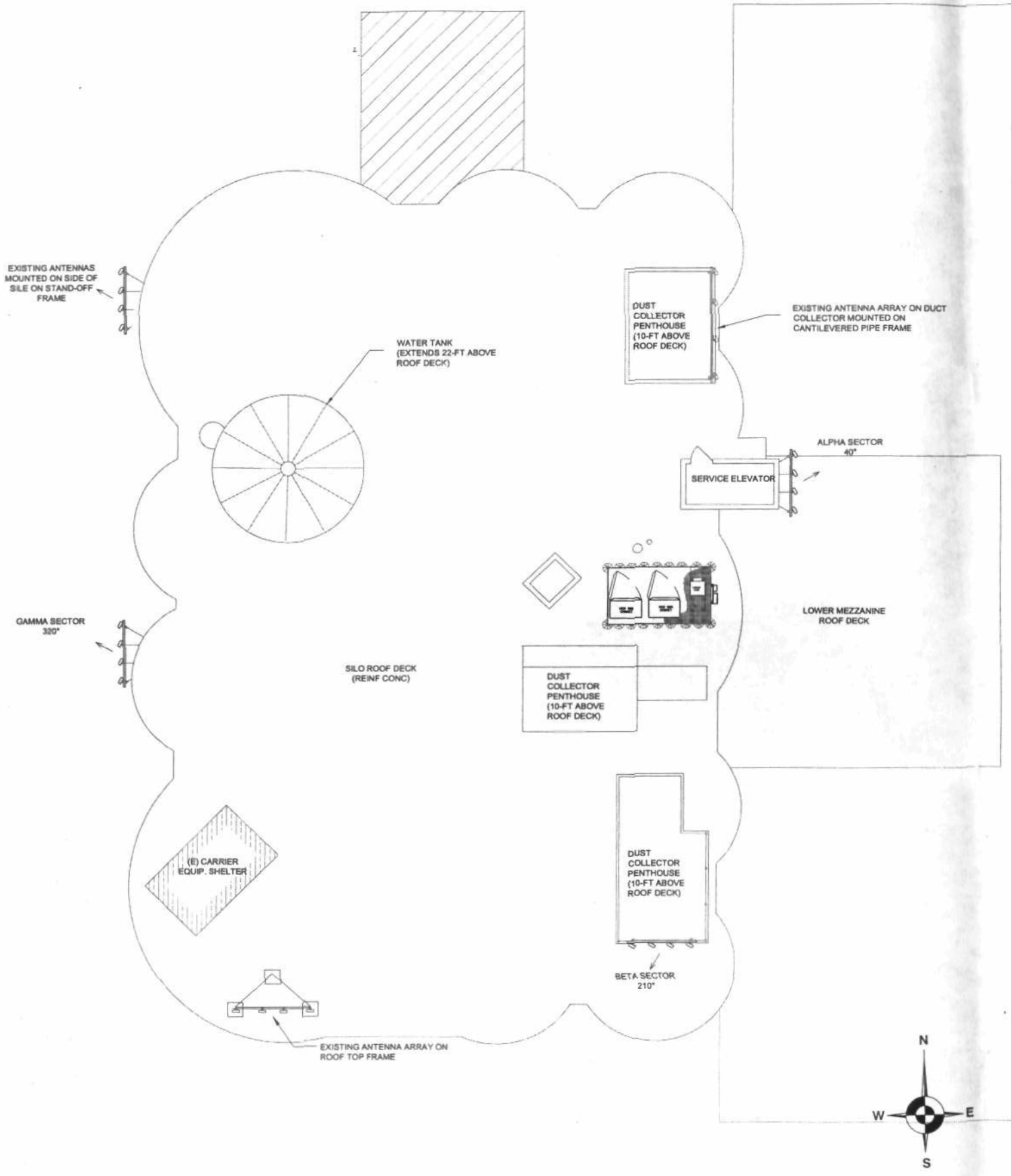
SITE NO.
1469

SITE NAME
LYONS

SITE ADDRESS
5145 UTE HWY
LYONS, CO

SHEET TITLE
SITE PLAN

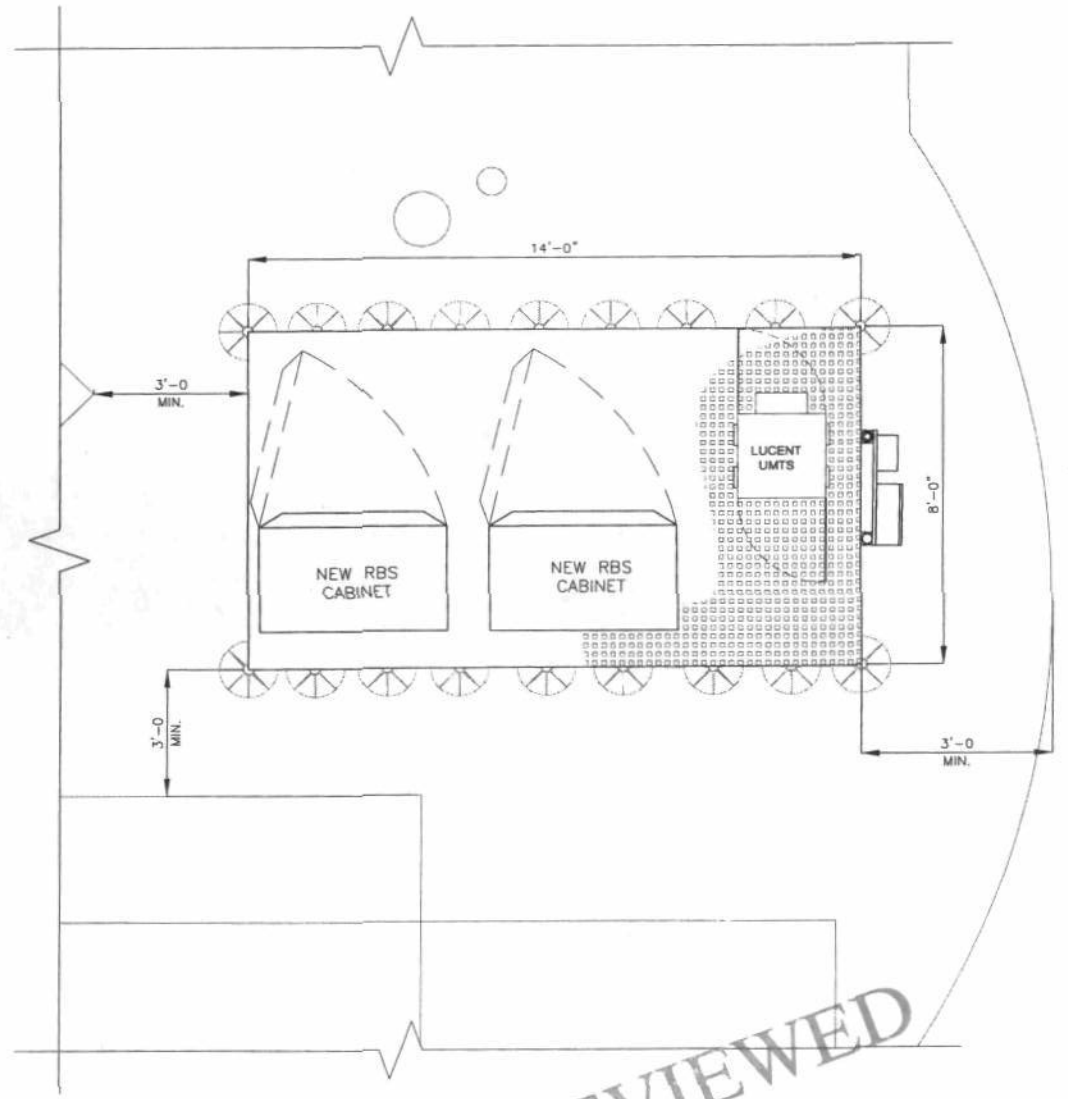
SHEET NO.
C-1
SHEET 2 OF 11



ENLARGED SITE PLAN

SCALE 1/8" = 1'-0"

1



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BOULDER COUNTY
BUILDING DIVISION

ENLARGED EQUIPMENT PLAN

SCALE 1/2" = 1'-0"

2

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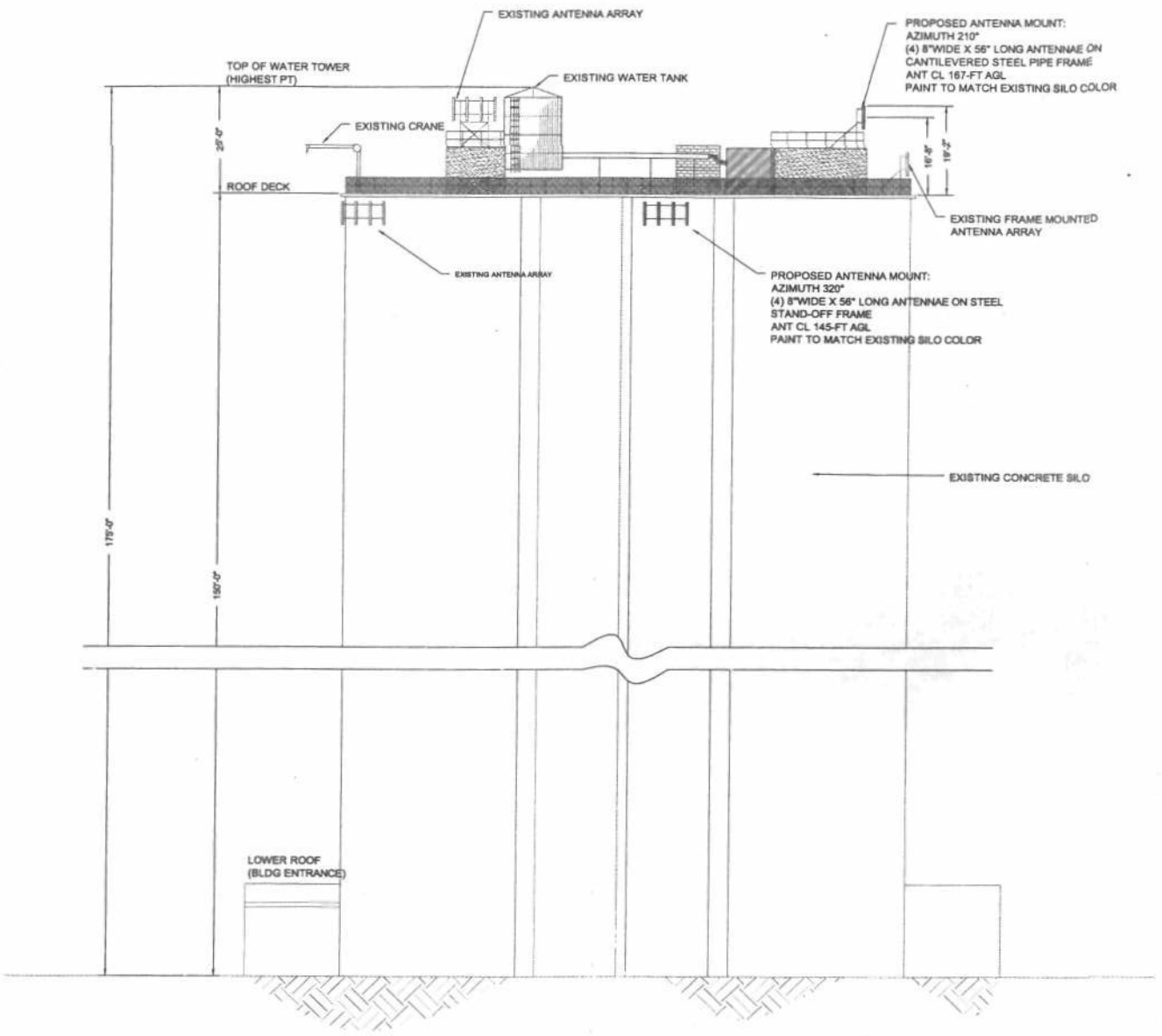
SITE NAME
LYONS

SITE ADDRESS
5145 UTE HWY
LYONS, CO

SHEET TITLE
ANTENNA & EQUIPMENT PLAN

SHEET NO.
C-2

SHEET 3 OF 11

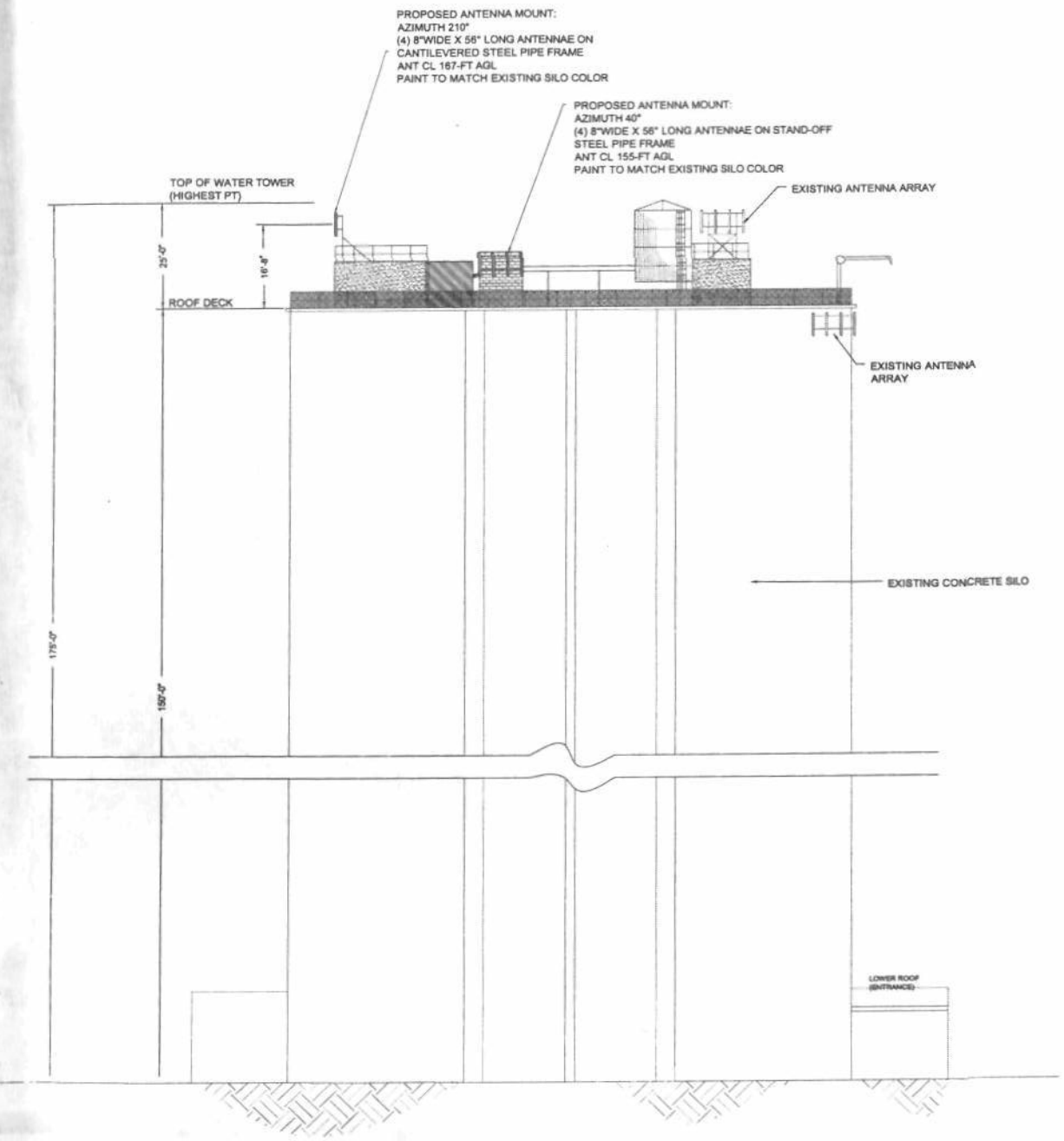


NOTE:

ALL DIMENSIONS SHOWN TO BE FIELD VERIFIED.
ELEVATIONS TO BE VERIFIED WITH FAA 1A
REPORT BY PLS.

WEST ELEVATION

SCALE
1/16" = 1'-0"



EAST ELEVATION

SCALE
1/16" = 1'-0"

* PLEASE READ THIS SECTION BEFORE INSTALLATION → VERTICAL CONDUCTOR SUPPORTS PER TABLE 300.19 (A) NEC

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1	5/8/07	CO PERMIT APPLICATION			

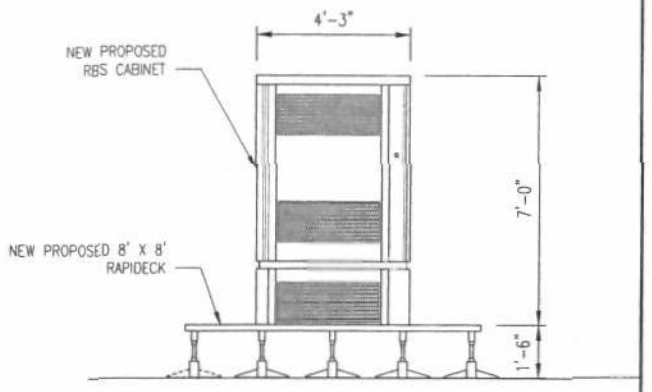
SITE NO.
1469

SITE NAME
LYONS

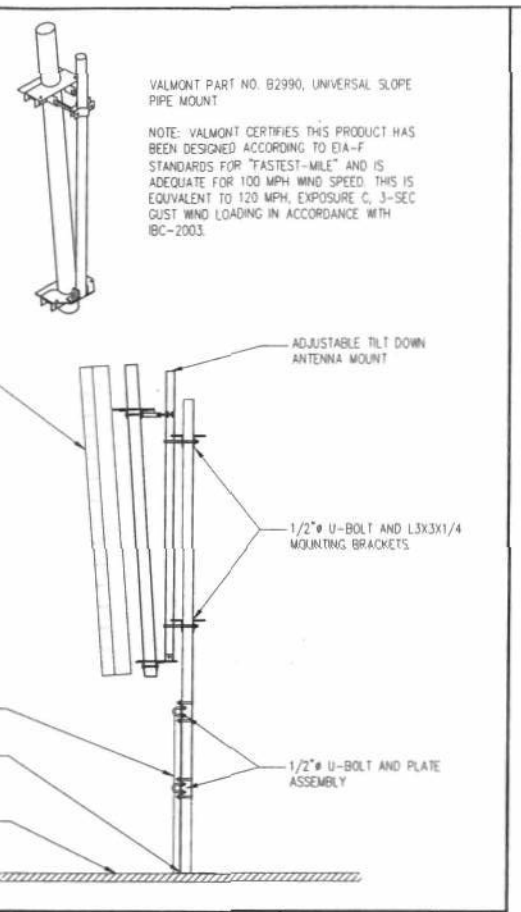
SITE ADDRESS
5145 UTE HWY
LYONS, CO

SHEET TITLE
EAST/WEST
ELEVATIONS

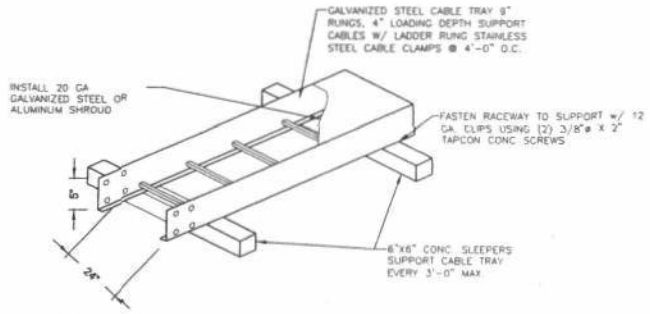
SHEET NO.
C-3
SHEET 4 OF 11



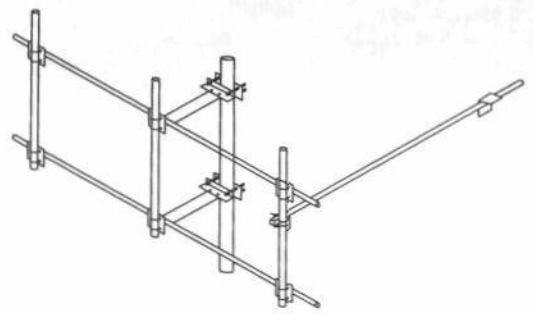
EQUIPMENT ELEVATION | SCALE: NTS | 1



ENLARGED SITE PLAN | SCALE: NTS | 2



CABLE TRAY | SCALE: NTS | 3



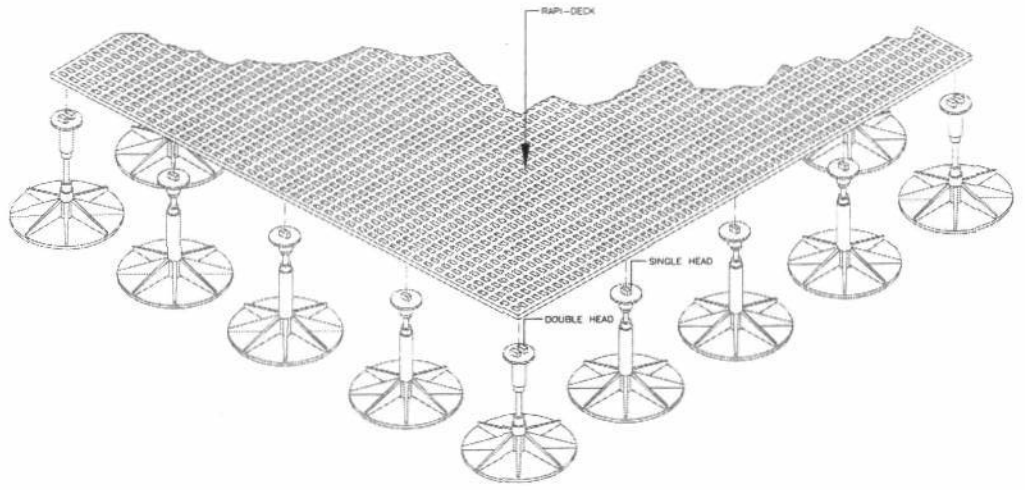
MTS PART SF-TP10-72
STAND-OFF PIPE FRAME

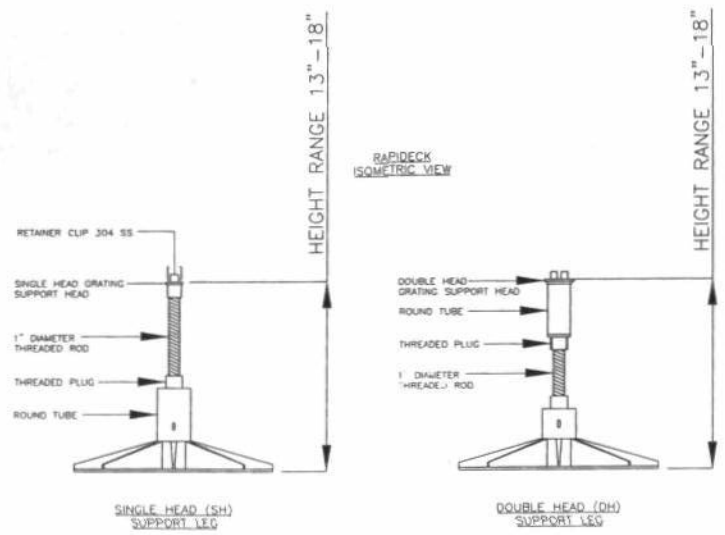
NOTE: THIS PRODUCT HAS BEEN DESIGNED ACCORDING TO EIA-F STANDARDS FOR "FASTEST-MILE" AND IS ADEQUATE FOR 100 MPH WIND SPEED. THIS IS EQUVALENT TO 120 MPH, EXPOSURE C, 3-SEC GUST WIND LOADING IN ACCORDANCE WITH IBC-2003.

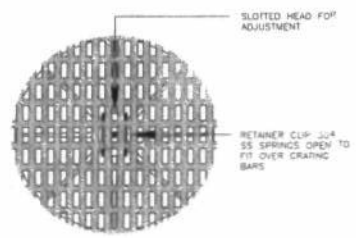
ATTACH BRACKET TO CONCRETE SILO w/ 1/2"ø EPOXY ANCHORS (IE: HILTI HVA OR EQUAL)

NOTE: EXPANSION ANCHORS NOT PERMITTED!

SCALE: NTS | 4









PRE-FABRICATED PLATFORM | SCALE: NTS | 5

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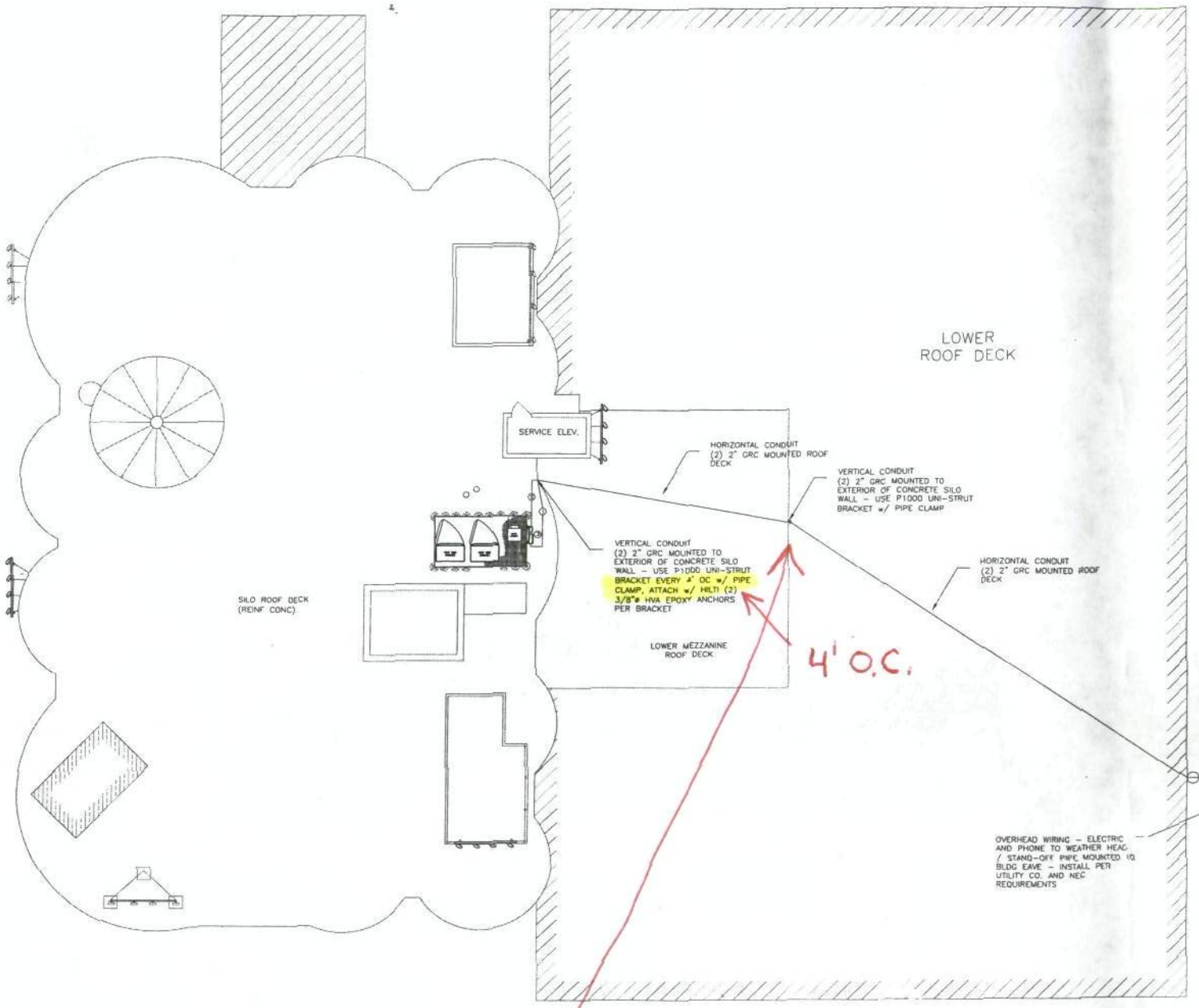
SITE NO.
1469

SITE NAME
LYONS

SITE ADDRESS
5145 UTE HWY
LYONS, CO

SHEET TITLE
CONSTRUCTION DETAILS

SHEET NO.
C-4
SHEET 5 OF 11



CONDUCTOR SUPPORTS ARE REQUIRED ACCORDING TO TABLE 300.19(A)

* PROVIDE MEASUREMENT TOOL/DEVICE. POSSIBLY WIRE PULLING TAPE WITH FEET MARKERS.

OVERHEAD CLEARANCE PER 225,60

26.5 ft FOR RAIL

GENERAL NOTES:

1. CONTRACTOR MUST VERIFY LOCATION OF ALL BURIED UTILITIES PRIOR TO ANY EXCAVATION.
2. COORDINATE WITH ELECTRICAL UTILITY COMPANY FOR EXACT METERING REQUIREMENTS AND SERVICE ROUTING. APPLY FOR ELECTRICAL SERVICE ASAP.
3. COORDINATE EXACT TELEPHONE REQUIREMENTS AND SERVICE ROUTING WITH LOCAL TELEPHONE COMPANY. APPLY FOR TELEPHONE SERVICE ASAP.
4. IT IS THE GENERAL CONTRACTOR'S RESPONSIBILITY TO APPLY FOR COMMERCIAL POWER IMMEDIATELY UPON RECEIVING AWARD OF BID. THE GENERAL CONTRACTOR IS REQUIRED TO KEEP ALL RECEIPTS FROM THE POWER COMPANY ACKNOWLEDGING APPLICATION FOR POWER AND THOROUGH DOCUMENTATION OF ANY DISCUSSIONS WITH THE POWER COMPANY THEREAFTER. ALSO, THE GENERAL CONTRACTOR SHALL RECEIVE IN WRITING A CONFIRMATION FROM THE POWER COMPANY AS TO WHEN THE ANTICIPATED POWER CONNECTION WILL BE COMPLETE. IF COMMERCIAL POWER IS NOT AVAILABLE BY THE "POWER COMPLETE" DATE AS CALLED OUT IN THE SPECIFICATIONS, A GENERATOR WILL BE SUPPLIED AND MAINTAINED BY THE GENERAL CONTRACTOR UNTIL COMMERCIAL POWER IS OBTAINED. ALL COSTS ASSOCIATED WITH THE GENERATOR WILL BE MUTUALLY AGREED UPON BETWEEN THE OWNER, ARCHITECT AND GENERAL CONTRACTOR. IN THE EVENT THE GENERAL CONTRACTOR FAILS TO TAKE THE NECESSARY MEASURES AS DESCRIBED HERE-IN TO SECURE POWER BY THE POWER COMPLETE DATE, THEN ALL COSTS ASSOCIATED WITH THE GENERATOR WILL BE BORNE BY THE GENERAL CONTRACTOR.

FLAG NOTES THIS SHEET:

① ELECTRICAL SERVICE ENTRANCE. VERIFY EXACT LOCATION AND ROUTING OF INCOMING SERVICE WITH UTILITY AND OWNER. FIELD VERIFY EXACT LOCATION OF UTILITY COMPANY TRANSFORMER AND PROVIDE BASE BID FOR CONDUIT LENGTH BASED ON FIELD SURVEY. ROUTE NEW SERVICE UNDERGROUND FROM EXISTING TRANSFORMER TO NEW METER AND DISCONNECT THEN TO NEW POWER PEDESTAL. PROVIDE GALVANIZED RIGID CONDUIT (GRC) FOR 30'-0" FROM POWER PEDESTAL. PROVIDE WEATHERPROOF PULL BOX FOR EVERY 270° (MAXIMUM) OF BENDS. COORDINATE EXACT REQUIREMENTS WITH UTILITY AND OWNER.

② TELEPHONE SERVICE ENTRANCE. VERIFY EXACT LOCATION AND ROUTING OF INCOMING SERVICE WITH UTILITY AND OWNER. ROUTE SERVICE UNDERGROUND FROM EXISTING TELCO PEDESTAL (FIELD VERIFY EXACT LOCATION) TO NEW POWER PEDESTAL. PROVIDE (2) 2" CONDUITS WITH PULL WIRE. PROVIDE GALVANIZED RIGID CONDUIT (GRC) FOR 30'-0" FROM POWER PEDESTAL. PROVIDE WEATHERPROOF PULL BOX FOR EVERY 360° (MAXIMUM) OF BENDS. COORDINATE EXACT REQUIREMENTS WITH UTILITY AND OWNER.

③ POWER PEDESTAL PROVIDED.

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BOULDER COUNTY
BUILDING DIVISION

NOTE: A PLAQUE IS REQUIRED. NEC 225.37 PLEASE READ AND INSTALL PER NEC

ELECTRICAL ROUTING PLAN

SCALE 3/32" = 1'-0"

1

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REV	DATE	DESCRIPTION	DRAWN BY	CHECKED BY	APP'D BY
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SITE NO.
1469

SITE NAME
LYONS

SITE ADDRESS
5145 UTE HWY
LYONS, CO

SHEET TITLE
ELEVATIONS

SHEET NO.
E-1
SHEET 6 OF 11

PANEL LA

120/240 VOLTS 1 PHASE 3 WIRES, SN 200 AMP BUS, 200 AMP MAIN, CB
10,000 A.I.C. RATING (MINIMUM). NEMA 3R ENCLOSURE, SURFACE MOUNTED

MFG RE: 1-LINE

DESCRIPTION	WIRE	BKR	POLE	CKT	VA	Φ	VA	CKT	POLE	BKR	WIRE	DESCRIPTION
TVSS	#6	60	2	1		A	1000	2	1	20	#12	LIGHTS
				3		B	3300	4	2	60	#6	NEW RBS #2106
NEW RBS #2106	#6	60	2	5	3300	A	3300	6				
				7	3300	B	360	8	1	20	#12	RECEPT
SPARE		20	1	9		A		10	1	20		SPARE
SPACE				11		B		12	1	20		SPARE
				13		A		14				SPACE
				15		B		16				
				17		A		18				
				19		B		20				

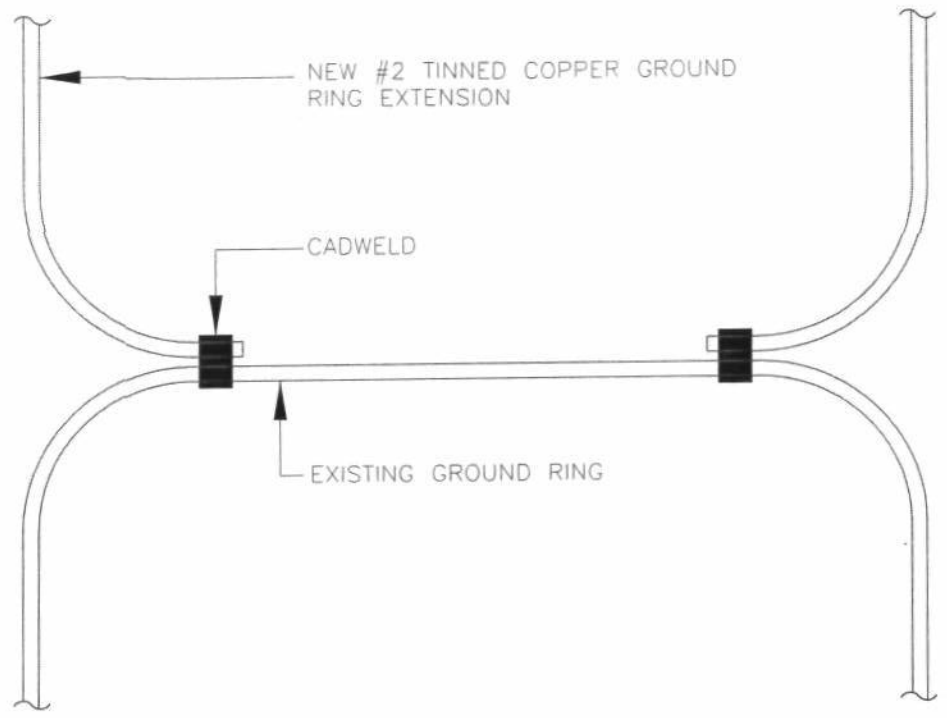
* CONTRACTOR TO PROVIDE CIRCUIT BREAKER

PANEL NOTE:
TVSS INSTALLATION PER ELECTRICAL CONTRACTOR

PANEL LOAD CALCULATIONS

LIGHTING	1000	VOLTAMPS X	1.25 =	1250	VOLTAMPS
RECEPTACLES - 1ST 10 KVA	360	VOLTAMPS X	1.00 =	230	VOLTAMPS
RBS(1+2)	13,200	VOLTAMPS X	1.00 =	13,200	VOLTAMPS
		VOLTAMPS X	1.25 =		VOLTAMPS
MISC. LOADS		VOLTAMPS X	1.00 =		VOLTAMPS
				14,810	TOTAL DEMAND VA
				61.7	AMPS, 1Ø, 3W

PANEL SCHEDULE — SCALE NTS — 1



GROUND EXTENSION — SCALE NTS — 2

ELECTRICAL NOTES:

1. ALL ELECTRICAL WORK SHALL BE PERFORMED IN ACCORDANCE WITH THE PROJECT SPECIFICATIONS AND ALL APPLICABLE LOCAL CODES.
2. CONDUIT ROUTINGS ARE SCHEMATIC. SUBCONTRACTOR SHALL INSTALL CONDUITS SO THAT ACCESS TO EQUIPMENT IS NOT BLOCKED.
3. THE SUBCONTRACTOR IS RESPONSIBLE FOR PROPERLY SEQUENCING GROUNDING AND UNDERGROUND CONDUIT INSTALLATION AS TO PREVENT ANY LOSS OF CONTINUITY IN THE GROUNDING SYSTEM OR DAMAGE TO THE CONDUIT.
4. COMPRESSION GROUND CONNECTIONS MAY BE REPLACED BY EXOTHERMIC (CADWELD) CONNECTIONS.
5. ALL GROUND CONNECTIONS BELOW GRADE SHALL BE EXOTHERMIC (CADWELD).
6. ALL GROUND CONNECTIONS ABOVE GRADE (INTERIOR & EXTERIOR) SHALL BE FORMED USING HIGH PRESS CRIMPS.
7. ALL EXOTHERMIC CONNECTIONS TO THE GROUND RODS SHALL START AT THE TOP & HAVE A VERTICAL SEPARATION OF 6" FOR EVERY ADDITIONAL CONNECTION.
8. ALL EXTERIOR GROUND CONNECTIONS SHALL BE COATED WITH A CORROSION RESISTANT MATERIAL.
9. ALL EXTERIOR GROUND CONDUCTORS SHALL BE #2 AWG SOLID TINNED COPPER UNLESS OTHERWISE INDICATED.
10. GROUND RODS SHALL BE COPPER CLAD STEEL, 5/8"ø 10-FT. LONG, AND SHALL BE DRIVEN VERTICALLY WITH THEIR TOPS 18" BELOW FINAL GRADE OR 6" BELOW FROST LINE FOR MAXIMUM DEPTH.
11. CONNECTIONS TO THE GROUND BUS SHALL NOT BE DOUBLED UP OR STACKED BACK TO BACK CONNECTIONS ON OPPOSITE SIDES OF THE GROUND BUS ARE PERMITTED.
12. USE OF 90° BENDS IN THE PROTECTION GROUNDING CONDUCTORS SHALL BE A VOIDED WHEN 45° BENDS CAN BE ADEQUATELY SUPPORTED.
13. MAXIMUM RESISTANCE OF THE COMPLETED GROUND SYSTEM SHALL NOT EXCEED 5 OHMS. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH PROJECT SPECIFICATION FOR FACILITY GROUNDING, USING FALL OF POTENTIAL METHOD.
14. BOND ALL METALLIC OBJECTS WITHIN 6FT OF MAIN GROUND WIRES WITH - #2 AWG TIN-PLATED COPPER GROUND CONDUCTOR.

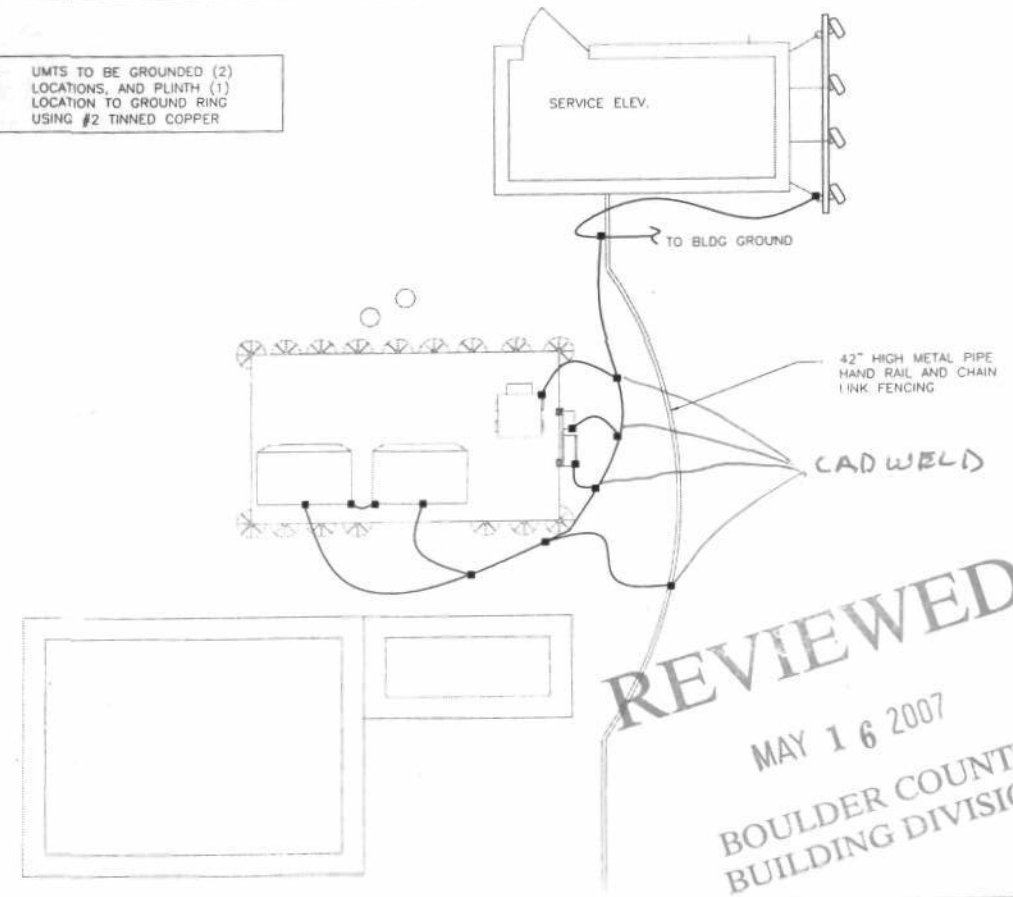
ELECTRICAL NOTES — SCALE NTS — 3

GROUNDING NOTES LEGEND:

1. ALL GROUNDING SHALL BE IAW NEC-2002 AND LOCAL CODES.
2. PROVIDE #2 AWG SOLID BARE TINNED COPPER CONDUCTOR TO BOND REMOTE POWER PEDESTALS GROUND BAR TO BURIED GROUND RING. PROVIDE EXOTHERMIC WELD AT BURIED GROUND RING.
3. PROVIDE #2 AWG SOLID BARE TINNED COPPER CONDUCTOR TO BOND METER TO SERVICE ENTRANCE GROUND ROD. PROVIDE EXOTHERMIC WELD AT GROUND ROD.
4. 5/8" x 10'-0" LONG COPPER CLAD GROUND ROD. SPACING BETWEEN RODS AS SHOWN. TYPICAL FOR ALL GROUND RODS, UNLESS NOTED OTHERWISE.
5. PROVIDE #2 AWG SOLID BARE TINNED COPPER GROUNDING CONDUCTOR TO BOND MASTER GROUND BAR TO BURIED GROUND RING. PROVIDE INSPECTION SLEEVE AT BURIED GROUND RING. SEE INSPECTION SLEEVE DETAIL.

GROUNDING NOTES — SCALE NTS — 4

NOTE: UMTS TO BE GROUNDED (2) LOCATIONS, AND PLINTH (1) LOCATION TO GROUND RING USING #2 TINNED COPPER



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PLATFORM GROUNDING PLAN — SCALE 1/4" = 1'-0" — 5

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REV	DATE	DESCRIPTION	Drawn BY	CHECKED BY	APP'D BY
4	4/30/07	CD/PERMIT APPLICATION			

SITE NO.
1469

SITE NAME
LYONS

SITE ADDRESS
5145 UTE HWY
LYONS, CO

SHEET TITLE
GROUNDING PLAN & DETAILS

SHEET NO.
E-2
SHEET 7 OF 11

1. THE ELECTRICAL CONTRACTOR SHALL COORDINATE ALL CONDUIT ROUTING IN FIELD WITH SITE CONSTRUCTION MANAGER.

2. ELECTRICAL AND TELEPHONE SERVICES SHOWN ARE PROPOSED. THE ELECTRICAL CONTRACTOR SHALL VERIFY EXISTING CONDITIONS AND COORDINATE CONNECTION POINTS, ROUTING, AND ASSOCIATED REQUIREMENTS IN FIELD.

3. ALL CONDUITS ROUTED BELOW GRADE SHALL BE PVC SCH 40. RISER ELBOWS TO BE PVC SCH 80 SWEEP WITH 36" MIN. RADIUS. CONDUIT RISERS ABOVE GRADE TO BE PVC SCH 80. *

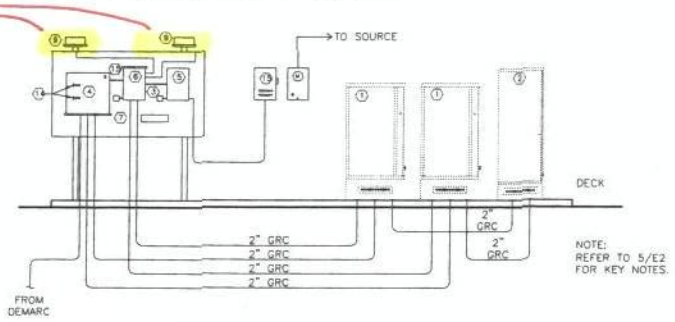
4. ELECTRICAL CONTRACTOR SHALL PROVIDE A 2" PVC CONDUIT WITH EXPANSION SLIP JOINT FROM 2" STUB-UPS ON BOTH ENDS OF TELCO/INTERFACE/ALRAM CONDUIT.

5. ELECTRICAL CONTRACTOR SHALL PROVIDE A 2" PVC CONDUIT WITH EXPANSION SLIP JOINT FROM 2" STUB-UPS ON BOTH ENDS OF SERVICE ENTRANCE CONDUIT.

6. CIRCUIT BREAKERS TO BE INDENTIFIED WITH TYPEWRITTEN OR MECHANICALLY PRINTED LABELS.

7. ALL 1" SEALTITE FOR GROUND WIRES TO BE SEALED.

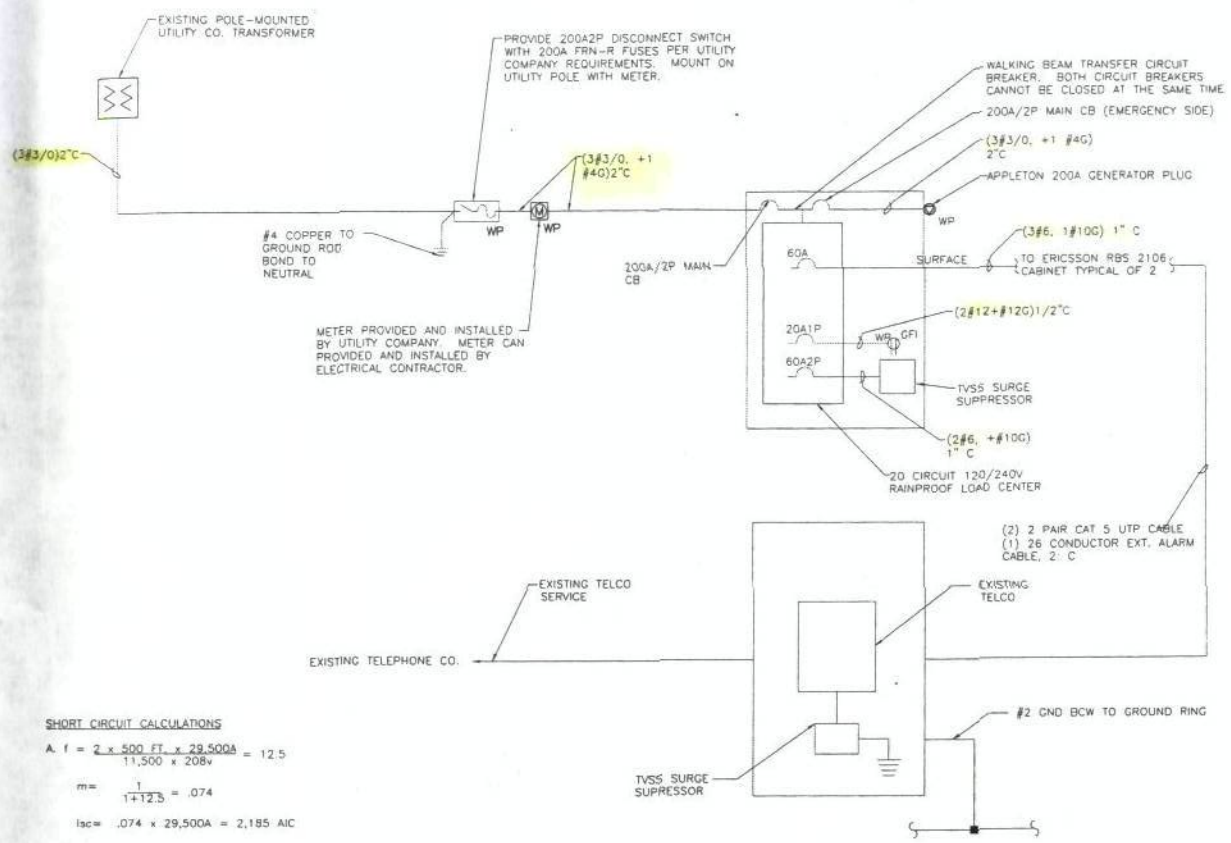
MUST BE DOWN LIT FIXTURE



KEY NOTES

①	ERICSSON RBS 2106 CABINET	⑧	(2) 4 PAIR CAT. 5 SHIELDED COPPER WIRE IN 1" PVC/SEALTITE
②	GRAYSON E911 CABINET	⑨	WEATHERPROOF HALOGEN LIGHTS OUTDOOR FIXTURE (2-500W, ONE FACING BOARD, ONE FACING PLATFORM) * MUST BE DOWN LIT FIXTURE
③	GENERATOR RECEPTACLE (APPLETON #AJA20033-200RS)	⑩	3-#6 AWG + 1-#8 GND IN 2" SEALTITE, LEAVE 6" SEALTITE WITH 8" OF ADDITIONAL CABLE LENGTH COILED ON PLATFORM.
④	COMMUNICATION BOX (HOFFMAN NEMA 3R 30" X 30" x 12" D)	⑪	(2) 6 PAIR CAT. 5 SHIELDED COPPER WIRE IN 2" PVC CONDUIT PROVIDED WEATHERPROOF IN GROUND PULLBOXES (SIZE AS REQUIRED) TO MEET N.E.C.
⑤	TRANSFER SWITCH (NON-FUSED) (CUTLER HAMMER MTVX 225)	⑫	METER BASE, 240V.
⑥	20 CIRCUIT 120/240V RAINPROOF LOAD CENTER W/200 AMP MAIN BREAKER OR APPROVED EQUAL (GE MODEL # TM2020RCU MOD. 2)	⑬	200 AMP./2P., 240V SERVICE DISCONNECT IN NEMA 3R ENCLOSURE FUSED AT 200 AMP
⑦	TVSS SURGE SUPPRESSOR (ATLANTIC SCIENTIFIC MOD. # 11215)	⑭	T-1 SURGE SUPPRESSOR (2) ATLANTIC SCIENTIFIC #90700
		⑮	125V/20A GFI DUPLEX RECEPTACLE IN OUTDOOR WHITE 2" x 4" BOX W/ HINGED COVER.

SCHEMATIC WIRING PLAN — SCALE NTS — 1



SHORT CIRCUIT CALCULATIONS

A. $f = \frac{2 \times 500\ FT. \times 29,500A}{11,500 \times 208v} = 12.5$

$m = \frac{1}{1+12.5} = .074$

Isc= .074 x 29,500A = 2,185 AIC

ONE LINE DIAGRAM — SCALE NTS — 2

NOTES:
1. TYPES AND SIZES OF THE ANTENNA CABLES ARE BASED ON THE ESTIMATED LENGTHS OF THE CABLES. CONTRACTOR SHALL VERIFY ACTUAL LENGTHS IN FIELD BEFORE INSTALLATION AND NOTIFY THE CONSTRUCTION MANAGER FOR CLARIFICATION.

2. ALL ANTENNA COAXIAL CABLES AND JUMPERS SHALL BE INSTALLED WITHOUT LOOPS AND / OR PIGTAILS.

3. COAXIAL CABLE GROUND KITS SHALL BE INSTALLED AS CLOSE TO THE CONNECTOR AS POSSIBLE AT EACH ANTENNA LOCATION.

4. ANTENNA COAXIAL CABLE GROUND KITS SHALL BE INSTALL AS CLOSE TO THE CONNECTOR AS POSSIBLE AT EACH ANTENNA LOCATION. IF THIS IS NOT POSSIBLE, THE GROUND KIT SHALL BE INSTALL IMMEDIATELY AFTER THE BEND ON THE FIRST RUN OF THE CABLE.

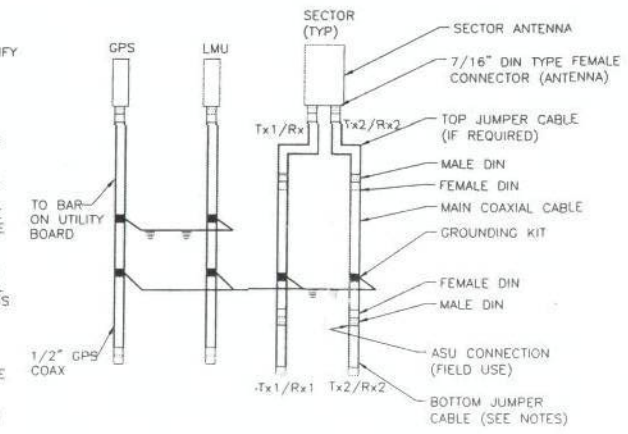
5. ANTENNA COAXIAL SHALL BE INSTALLED TO COMPLY WITH THE MANUFACTURERS MINIMUM BEND RADIUS. THE CONTRACTOR SHALL INSTALL RACEWAY FOR COAXIAL CABLE USING THE PROPER FITTINGS NECESSARY TO ENSURE THAT THE MINIMUM BEND RADIUS REQUIREMENTS ARE MET.

6. THE GPS ANTENNA COAXIAL CABLE SHALL BE A CONTINUOUS CABLE RUN FROM THE CONNECTOR AT THE ANTENNA HEAD TO THE CONNNECTOR AT THE CABINET, WITHOUT JUMPERS.

7. ALL ANTENNA COAXIAL CABLE SHALL BE MARKED AND TAGGED WITH THE PROPER COLOR CODE.

8. COLOR CODE AT ALL MAIN CABLES AT TWO LOCATIONS:
   - NEAR THE ANTENNA
   - NEAR THE EQUIPMENT

9. COAX GROUND KITS, COAX WEATHER PROOFING, SNAP-IN HANGER CLAMPS AND HOISTING GRIPS SHALL BE PROVIDED BY THE CONTRACTOR FOR INSTALLATION.

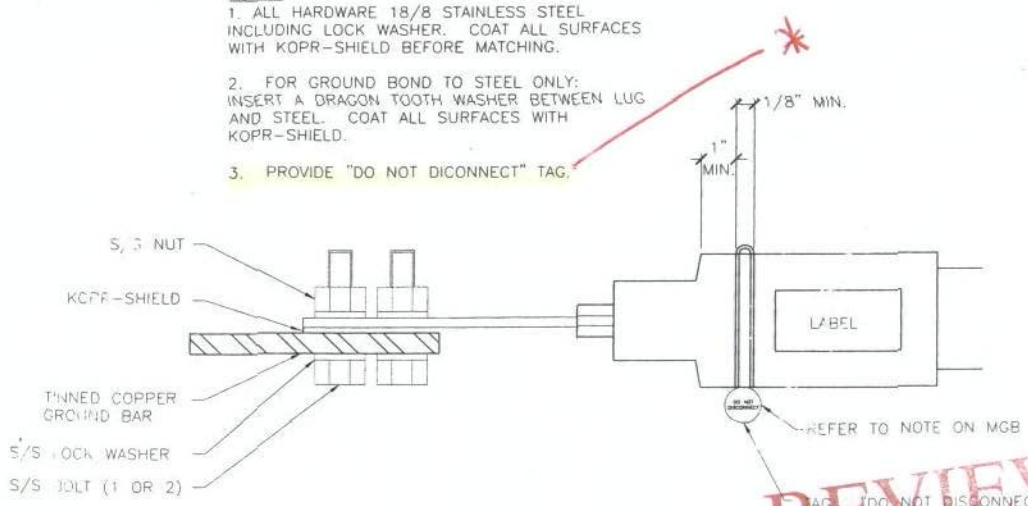


ANTENNA WIRING — SCALE NTS — 3

NOTES:
1. ALL HARDWARE 18/8 STAINLESS STEEL INCLUDING LOCK WASHER. COAT ALL SURFACES WITH KOPR-SHIELD BEFORE MATCHING.

2. FOR GROUND BOND TO STEEL ONLY: INSERT A DRAGON TOOTH WASHER BETWEEN LUG AND STEEL. COAT ALL SURFACES WITH KOPR-SHIELD.

3. PROVIDE "DO NOT DICONNECT" TAG. *



ANT. GROUNDING CONNECTOR — SCALE NTS — 5

REVIEWED
MAY 16 2007
BOULDER COUNTY BUILDING DIVISION

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COLORADO REGISTERED PROFESSIONAL ENGINEER
CHRISTOPHER PORCELLO
35670
2007 PE
5-1-07

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REV	DATE	DESCRIPTION	DRAWN BY	CHECKED BY	APP'D BY
4	5/2/07	CD PERMIT APPLICATION			

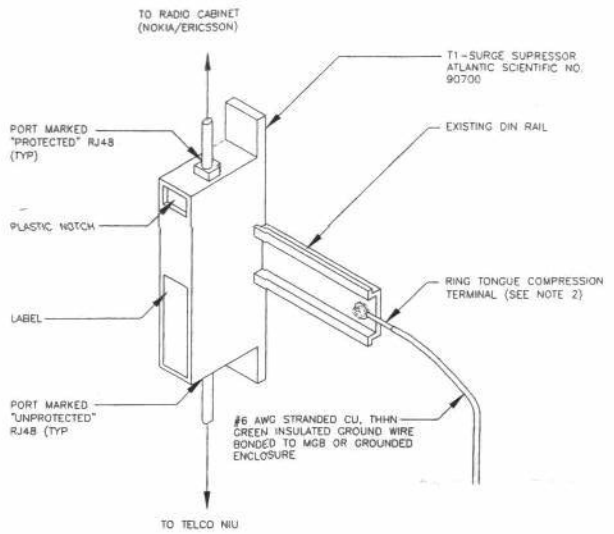
SITE NO.
1469

SITE NAME
LYONS

SITE ADDRESS
5145 UTE HWY
LYONS, CO

SHEET TITLE
ELECTRICAL DETAILS

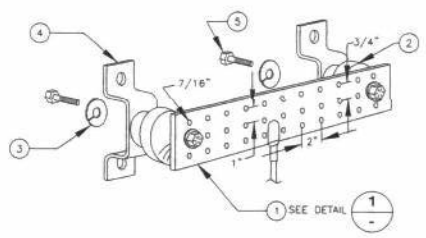
SHEET NO.
E-3
SHEET 8 OF 11



NOTES:

1. MOUNT SURGE ARRESTOR ON EXISTING DIN RAIL IF AVAILABLE, OTHERWISE PROVIDE ATLANTIC SCIENTIFIC MODEL NO. 21605, 5-INCH 4-UNIT DIN RAIL, OR EQUAL, AND SECURELY MOUNT TO TELCO BOARD.
2. ATTACH GROUND WIRE TO THE DIN RAIL, SEPARATELY FROM THE RAIL MOUNTING HARDWARE, USING STAINLESS STEEL BOLT, LOCKING WASHER AND NUT OR SELF-TAPPING MACHINE SCREW THAT FIRMLY GRIPS THE DIN RAIL.
3. TVSS MAY BE ORIENTED TO SUIT SITE REQUIREMENTS.

T-1 SURGE SUPRESSOR | 1



LEGEND

1- COPPER GROUND BAR, 1/4"X 4"X 20", NEWTON INSTRUMENT CO. CAT. NO. B-6142 OR EQUAL. HOLE CENTERS TO MATCH NEMA DOUBLE LUG CONFIGURATION. (ACTUAL GROUND BAR SIZE WILL VARY BASED ON NUMBER OF GROUND CONNECTIONS)

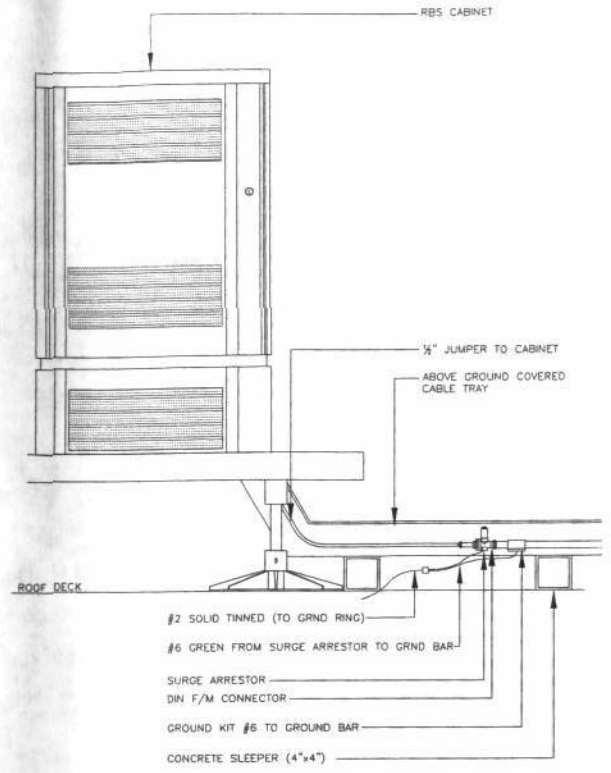
2- INSULATORS, NEWTON INSTRUMENT CAT. NO. 3061-4 OR EQUAL INSULATORS SHALL BE ELIMINATED WHEN BONDING DIRECTLY TO TOWER STRUCTURE. CONNECTION TO TOWER STRUCTURE SHALL BE PER MANUFACTURERS RECOMMENDATIONS.

3- 5/8" LOCKWASHERS, NEWTON INSTRUMENT CO. CAT. NO. 3015-8 OR EQUAL

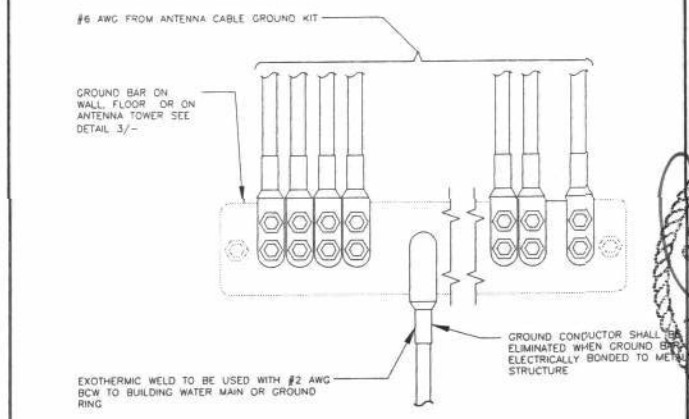
4- WALL MOUNTING BRACKET, NEWTON INSTRUMENT CO. CAT NO. A-6056 OR EQUAL

5- 5/8-11 X 1" HHCS BOLTS, NEWTON INSTRUMENT CO. CAT NO. 3012-1 OR EQUAL

GROUNDING BAR DETAIL | 2



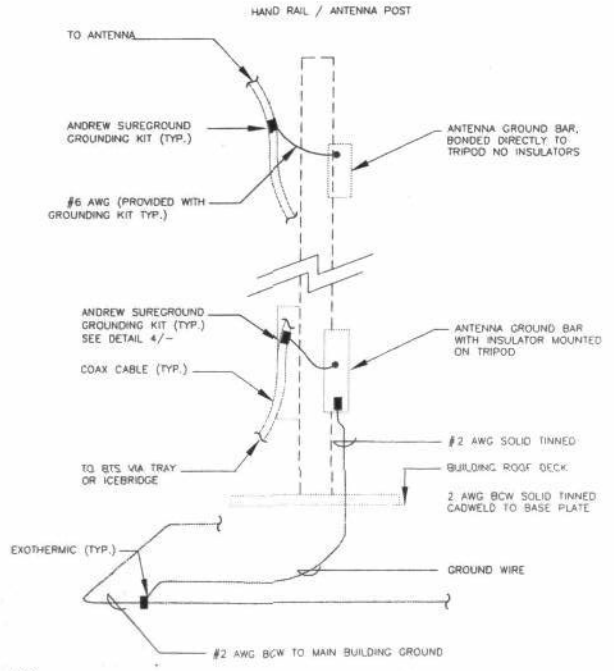
COAX/GROUND CONNECTION (RAPIDECK) | 3



NOTE: FOR SLOTED HOLES USE 3/8" FLAT WASHERS ON BOTH SIDES AND A LOCK WASHER ON THE NUT SIDE. FOR NON-SLOTTED HOLES USE JUST LOCK WASHER ON NUT SIDE.

*- GROUND BARS AT THE BOTTOM OF TOWERS SHALL ONLY USE EXOTHERMIC WELDS.

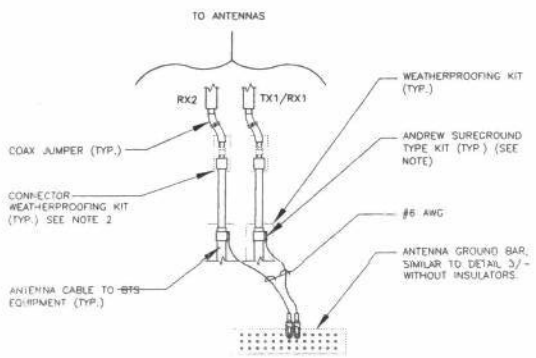
GRND WIRE / GRND BAR CONNECTION | 4



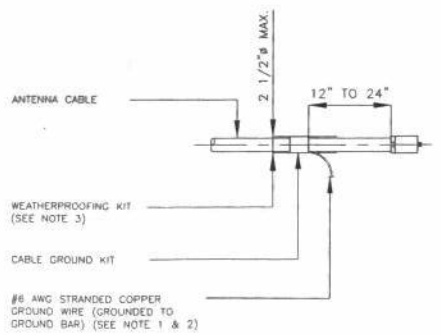
NOTE:
NUMBER OF GROUNDING BARS MAY VARY DEPENDING ON THE ANTENNA LOCATION AND CONNECTION ORIENTATION. PROVIDE AS REQUIRED

GROUNDING BAR CONNECTION | 5

NOTES:
1. DO NOT INSTALL CABLE GROUND KIT AT A BEND AND ALWAYS DIRECT GROUND WIRE DOWN TO ANTENNA GROUND BAR.
2. WEATHER PROOFING SHALL BE ANDREW TWO-PART TAPE KIT. COLD SHRINK SHALL NOT BE USED.



GROUNDING BAR CONNECTION | 6



NOTES:
1. DO NOT INSTALL CABLE GROUND KIT AT A BEND AND ALWAYS DIRECT GROUND WIRE DOWN TO GROUND BAR.
2. GROUNDING KIT SHALL BE ANDREW SUREGROUND TYPE KIT WITH TWO-HOLE LUG.
3. WEATHER PROOFING SHALL BE ANDREW TWO-PART TAPE SUPPLIED WITH KIT. COLD SHRINK SHALL NOT BE USED.

CABLE GROUND KIT DETAIL | 7

ABBREVIATIONS:

AGB	ANTENNA GROUND BAR
AWG	AMERICAN WIRE GAUGE
BCW	BARE COPPER WIRE
DWG	DRAWING
EMT	ELECTRICAL METALLIC TUBING
GEN	GENERATOR
IGR	INTERIOR GROUND RING (HALO)
IMC	INTERMEDIATE METALLIC CONDUIT
MGB	MASTER GROUND BAR
PCS	PERSONAL COMMUNICATION SYSTEM
PTS	POWER TRANSFER SWITCH
PVC	RIGID (SCH. 40 U.N.O.) POLYVINYL CHLORIDE CONDUIT
RGS	RIGID GALVANIZED STEEL
RWY	RACEWAY
TYP	TYPICAL

ELECTRICAL SYMBOLS:

- CHEMICAL GROUND ROD
- GROUND ROD
- DISCONNECT SWITCH
- METER
- CIRCUIT BREAKER
- CADWELD TYPE CONNECTION
- COMPRESSION TYPE CONNECTION
- GROUNDING WIRE
- XXX / XXX — REPRESENTS DETAIL NUMBER / REF. DRAWING NUMBER

SYMBOLS | 8

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BUILDING DIVISION

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REV	DATE	DESCRIPTION	DRAWN BY	CHECKED BY	APP'D BY
A	4/20/07	CD PERMIT APPLICATION			

SITE NO.
1469

SITE NAME
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SITE ADDRESS
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SHEET TITLE
ELECTRICAL & GROUNDING DETAILS

SHEET NO.
E-4
SHEET 9 OF 11

CONTRACTOR GENERAL NOTES

1. CONTRACTOR WILL PROVIDE ALL LABOR, MATERIALS, TOOLS, EQUIPMENT, SAFETY EQUIPMENT, TRANSPORTATION AND SERVICES NECESSARY TO COMPLETE ALL THE WORK OUTLINED IN ALL THE DRAWINGS, SPECIFICATIONS, SCOPE OF WORK, BILL OF MATERIALS AND ANY OTHER DOCUMENT ISSUED BY OWNER WIRELESS SERVICES.

2. THE CONTRACTOR SHALL COMPLY WITH ALL LOCAL AND NATIONAL CODES, LAWS, ORDINANCES, REGULATIONS, SAFETY REGULATIONS, ALL OSHA REGULATIONS, ALL PUBLIC AND MUNICIPAL AUTHORITIES AND ANY UTILITY COMPANY REGULATIONS AND DIRECTIVES.

3. ALL MATERIALS SUPPLIED BY THE OWNER REPRESENTATIVE AND THE CONTRACTOR, SHALL BE INSTALLED IN ACCORDANCE WITH ALL LOCAL AND NATIONAL CODES, LAWS, ORDINANCES, REGULATIONS AND PER MANUFACTURER'S RECOMMENDATIONS.

4. ANY CONTRACTOR SUBMITTING BIDS ON ANY OF THE WORK IS REQUIRED TO VISIT EACH SITE PRIOR TO THE BID SUBMITTAL AND FAMILIARIZE HIMSELF/HERSELF WITH THE EXISTING CONDITIONS AND UNDERSTAND THE SCOPE OF WORK INTENDED FOR THE PROJECT. THIS WILL BE PERFORMED AT THE CONTRACTOR'S EXPENSE.

5. THE DRAWINGS AND SPECIFICATIONS ARE A GENERAL DIRECTIVE FOR THE SCOPE OF WORK. EXACT DIMENSIONS AND LOCATIONS MAY CHANGE IN THE FIELD. THE CONTRACTOR IS TO VERIFY THE DIMENSIONS AND LOCATIONS AND REPORT ANY AND ALL DISCREPANCIES TO OWNER PRIOR TO COMMENCING THE RELATED WORK. ANY MINOR ERRORS OR OMISSIONS IN THE DRAWINGS AND SPECIFICATIONS DO NOT EXCUSE THE CONTRACTOR FROM COMPLETING THE PROJECT AND IMPROVEMENTS IN ACCORDANCE WITH THE INTENT OF THESE DOCUMENTS.

6. ALL SITES SHALL BE KEPT CLEAN AND FREE OF DEBRIS ON A DAILY BASIS. ALL TRASH AND MATERIAL NO LONGER BEING USED AT THE SITE MUST BE REMOVED AND PROPERLY DISPOSED OF ON A DAILY BASIS. ANY EXCESS MATERIAL WILL BE RETURNED TO OWNER AND DELIVERED TO THE WAREHOUSE FACILITY PER THE DIRECTION OF GENERAL DYNAMICS.

7. THE CONTRACTOR SHALL NOT PROCEED WITH ANY WORK AT THE SITE PRIOR TO A NOTICE TO PROCEED (NTP) HAS BEEN ISSUED AND A PRE-CONSTRUCTION MEETING AT THE SITE HAS TAKEN PLACE.

8. ALL WORK AND MATERIALS FURNISHED BY THE CONTRACTOR SHALL HAVE A WRITTEN 18 MONTH WARRANTY STARTING AT THE ACCEPTANCE OF THE SITE FROM OWNER.

9. THE CONTRACTOR SHALL HAVE A DESIGNATED MANAGER ON SITE AT ALL TIMES THAT ANY WORK IS BEING PERFORMED. A SUB-CONTRACTOR IS NOT DEFINED AS A DESIGNATED MANAGER.

10. THE CONTRACTOR SHALL PROVIDE A SCHEDULED LIST OF ALL SUB-CONTRACTORS, WHICH SHALL INCLUDE ADDRESSES, CELL PHONE NUMBERS, HOME PHONE NUMBERS, VERIFICATION OF INSURANCE, ANY AND ALL PERTINENT LICENSES AND AN ACCURATE SCHEDULE FOR THE PROJECT PRIOR TO THE ISSUANCE OF AN NTP.

11. THE CONTRACTOR IS TO KEEP A COMPLETE AND UP-TO-DATE SET OF THE DRAWINGS, SPECIFICATIONS, AND SCOPE OF WORK AND A BILL OF MATERIAL ON THE SITE AT ALL TIMES. THIS WILL BE REFERENCED AS THE AS-BUILT DRAWINGS AND MUST BE KEPT CURRENT ON A DAILY BASIS. THIS IS IN ADDITION TO THE PERMIT SET.

12. A NEW AND CLEAN SET OF CONTRACT DOCUMENTS WILL BE ISSUED TO THE CONTRACTOR NEAR THE COMPLETION OF THE PROJECT TO BE USED TO TRANSFER THE INFORMATION FROM THE FIELD COPY OF AS-BUILT DRAWING TO THE NEW COPY.

13. THE CONTRACTOR IS RESPONSIBLE FOR SCHEDULING ALL INSPECTIONS AND TESTING REQUIRED EACH PROJECT.

14. THE CONTRACTOR IS RESPONSIBLE TO FURNISH PROPER FACILITIES FOR THE WORKERS ON EACH PROJECT FOR THE DURATION OF THAT PROJECT.

15. THE CONTRACTOR IS RESPONSIBLE TO MAINTAIN THE PRESENT CONDITION OF ANY EXISTING BUILDINGS, LANDSCAPES, FENCING, EQUIPMENT, WALKS, DRIVES AND ATTACHMENTS. IF ANY DAMAGE SHOULD OCCUR, THE CONTRACTOR IS RESPONSIBLE TO RESTORE THE DAMAGE TO A BETTER OR NEW CONDITION.

PART 1 – PROJECT GENERAL NOTES

1.1 SCOPE OF WORKE OF W

A. PROVIDE ALL LABOR, MATERIAL, TOOLS, EQUIPMENT, TRANSPORTATION AND SERVICES NECESSARY FOR AND INCIDENTAL TO COMPLETION OF ALL WORK AS INDICATED ON THE DRAWINGS AND/OR AS SPECIFIED HEREIN.

1.2 DRAWING USE AND INTERPRETATION

A. THE DRAWINGS ARE DIAGRAMMATIC AND INDICATE THE GENERAL ARRANGEMENT OF SYSTEMS AND EQUIPMENT UNLESS INDICATED OTHERWISE BY DIMENSIONS OR DETAILS. EXACT EQUIPMENT LOCATIONS AND ROUTINGS, ETC. SHALL BE GOVERNED BY ACTUAL FIELD CONDITIONS AND/OR INSTRUCTIONS OF THE ENGINEER.

1.3 COMPLETE SYSTEMS

A. GENERAL: FURNISH AND INSTALL ALL MATERIALS AS REQUIRED FOR COMPLETE SYSTEMS, WHETHER SPECIFICALLY INDICATED OR NOT. ALL SYSTEMS SHALL BE COMPLETELY ASSEMBLED, TESTED, ADJUSTED, AND DEMONSTRATED TO BE READY FOR OPERATION PRIOR TO ACCEPTANCE.

1.4 CODES AND REGULATIONS

A. GENERAL: COMPLY WITH ALL GOVERNING FEDERAL, STATE AND LOCAL LAWS, ORDINANCES, CODES, RULES AND REGULATIONS. WHERE THE CONTRACT DOCUMENTS EXCEED THESE REQUIREMENTS, THE CONTRACT DOCUMENTS SHALL GOVERN. IN NO CASE SHALL WORK BE INSTALLED CONTRARY Y TO OR BELOW MINIMUM LEGAL STANDARDS.

B. UTILITIES: COMPLY WITH ALL APPLICABLE RULES, RESTRICTIONS AND REQUIREMENTS OF THE UTILITY COMPANIES SERVING THE PROJECT SITE/FACILITIES.

C. NON-COMPLIANCE: SHOULD ANY WORK BE PERFORMED WHICH IS FOUND NOT TO COMPLY WITH ANY OF THE ABOVE, CONTRACTOR SHALL PROVIDE ALL WORK AND PAY ALL COSTS NECESSARY TO CORRECT THE DEFICIENCIES.

1.5 REFERENCE STANDARDS

A. ALL LATEST PUBLISHED STANDARDS OF THE FOLLOWING ASSOCIATIONS/ORGANIZATIONS SHALL BE FOLLOWED AND APPLIED WHERE APPLICABLE, AS MINIMUM REQUIREMENTS.

1) (ANSI) AMERICAN NATIONAL STANDARDS INSTITUTE.
2) (ASTM) AMERICAN SOCIETY FOR TESTING AND MATERIALS.
3) (ETL) ELECTRICAL TESTING LABORATORY.
4) (ICEA) INSULATED CABLE ENGINEERS ASSOCIATION.
5) (IEEE) INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS.
6) (NBFU) NATIONAL BOARD OF FIRE UNDERWRITERS.
7) (NEMA) NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION.
8) (NECS) NATIONAL ELECTRIC SAFETY CODE.
9) (NFPA) NATIONAL FIRE PROTECTION ASSOCIATION.
10) (UL) UNDERWRITER'S LABORATORIES.
11) (BOCA) BUILDING OFFICIALS & CODE ADMINISTRATOR.
12) (NEC) NATIONAL ELECTRICAL CODE.

1.6 PERMITS

A. GENERAL: ALL PERMITS REQUIRED BY ALL APPLICABLE AGENCIES WILL BE OBTAINED BY THE CONTRACTOR.

1.7 SUBMITTALS

A. REQUIRED SUBMITTALS INCLUDE: LIST OF SUBCONTRACTORS; PRODUCT DATA; SHOP DRAWINGS; SAMPLES; TEST REPORTS; CERTIFICATIONS; WARRANTIES; MAINTENANCE MANUALS; RECORD DRAWINGS; AND VARIOUS ADMINISTRATIVE SUBMITTALS.

B. RECORD DOCUMENTS: INDICATE ACTUAL INSTALLED LOCATIONS FOR ALL EQUIPMENT AND DEVICES, ROUTING OF MAJOR INTERIOR RACEWAYS, LOCATIONS OF ALL CONCEALED AND UNDERGROUND EQUIPMENT AND RACEWAYS AND ALL APPROVED MODIFICATIONS TO THE CONTRACT DOCUMENTS AND DEVIATIONS NECESSITATED BY FIELD CONDITIONS AND CHANGE ORDERS.

C. SAFETY MEETING(S) MINUTES.

D. NUMBER OF COPIES OF ALL SUBMITTALS: TWO.

1.8 QUALITY ASSURANCE

A. MANUFACTURERS QUALIFICATIONS: NOT LESS THAN THREE YEARS EXPERIENCE IN THE ACTUAL PRODUCTION OF SPECIFIED PRODUCTS.

B. INSTALLERS QUALIFICATIONS: FIRM WITH NOT LESS THAN FIVE YEARS EXPERIENCE IN THE INSTALLATION OF ELECTRICAL SYSTEMS AND EQUIPMENT SIMILAR IN SCOPE AND COMPLEXITY TO THOSE REQUIRED FOR THIS PROJECT AND HAVING SUCCESSFULLY COMPLETED AT LEAST TEN COMPARABLE SCALE PROJECTS.

C. INCIDENTAL WORK: PAINTING, PATCHING, CARPENTRY, MECHANICAL WORK AND THE LIKE REQUIRED FOR SHALL BE PERFORMED BY CRAFTSMAN SKILLED IN THE APPROPRIATE TRADE.

1.9 INSPECTIONS

A. GENERAL: DURING AND UPON COMPLETION OF THE WORK, ARRANGE AND PAY ALL ASSOCIATED COSTS FOR INSPECTIONS OF ALL ELECTRICAL WORK INSTALLED UNDER THIS CONTRACT, IN ACCORDANCE WITH THE CONDITIONS OF THE CONTRACT.

B. INSPECTIONS REQUIRED: AS PER THE LAWS AND REGULATIONS OF THE LOCAL AND/OR STATE AGENCIES HAVING JURISDICTION AT THE PROJECT SITE.

C. INSPECTION AGENCY: APPROVED BY THE LOCAL AND/OR STATE AGENCIES HAVING JURISDICTION AT THE PROJECT SITE.

D. CERTIFICATES: SUBMIT ALL REQUIRED INSPECTION CERTIFICATES.

1.10 DELIVERY STORAGE AND HANDLING

A. PACKING AND SHIPPING: DELIVER PRODUCTS IN ORIGINAL UNOPENED PACKAGING, PROPERLY IDENTIFIED WITH MANUFACTURER'S IDENTIFICATION AND COMPLIANCE.

B. STORAGE AND PROTECTION: COMPLY WITH ALL MANUFACTURER'S WRITTEN RECOMMENDATIONS. STORE ALL PRODUCTS IN A MANNER WHICH SHALL PROTECT THEM FROM DAMAGE, WEATHER AND ENTRY OF DEBRIS.

C. DAMAGED PRODUCTS: DO NOT INSTALL DAMAGED PRODUCTS. ARRANGE FOR PROMPT REPLACEMENT.

1.11 MAINTAINING SITE CONDITIONS

A. THE CONTRACTOR IS RESPONSIBLE FOR MAINTAINING A NEAT AND ORDERLY PROJECT SITE. REMOVE AND DISPOSE OFF SITE ALL RUBBISH, WASTE, LITTER AND ALL FOREIGN SUBSTANCES IN A LEGAL MANNER. REMOVE PETRO-CHEMICAL SPILLS, STAINS AND OTHER FOREIGN DEPOSITS IN COMPLIANCE WITH OSHA REGULATIONS, RETURN ALL SURFACES TO ORIGINAL CONDITION.

LANDSCAPING

1. THE CONTRACTOR SHALL GRUB THE SITE AREA AND ANY ACCESS ROAD, CLEARING AND REMOVING A MINIMUM OF 6" TO ACHIEVE A STABLE SUB-BASE TO ACCEPT FILL OR OTHER MATERIAL SPECIFIED FOR THE SITE AND NOT REUSED IN ANY PART OF THIS PROJECT.

2. IT IS THE CONTRACTOR'S SOLE RESPONSIBILITY TO CALL THE LOCATING AUTHORITIES TO VERIFY THE LOCATION OF ANY UNDERGROUND UTILITIES OR LINES THAT EXIST WITHIN THE ENTIRE PROJECT AREA. THE CONTRACTOR SHALL PLACE THESE ITEMS ON THE AS-BUILT DRAWINGS.

3. IF LANDSCAPING IS REQUIRED, SUBMIT TWO COPIES OF THE LANDSCAPE PLAN UNDER NURSERY LETTERHEAD. IF A LANDSCAPE ALLOWANCE IS INCLUDED, PROVIDE A LIST OF PROPOSED ITEMS AS WELL AS A LANDSCAPE PLAN ON NURSERY LETTERHEAD.

4. ALL ROADWORK MATERIALS SHALL CONFORM TO ALL STATE AND LOCAL CODES IN ACCORDANCE WITH THE DEPARTMENT OF HIGHWAY AND PUBLIC TRANSPORTATION STANDARD SPECIFICATIONS.

5. THE ACCESS ROAD, IF REQUIRED, WILL BE CONSTRUCTED FIRST BEFORE ANY OTHER WORK ON THE SITE IS PERMITTED. BRING THE ROAD TO SUB BASE COURSE TO ALLOW CONSTRUCTION TRAFFIC TO USE THE ROAD FOR THE PROJECT AND COMPLETE THE ROAD AFTER THE CONSTRUCTION OF THE SITE IS SUBSTANTIALLY COMPLETE.

6. ALL SITE AREAS AND ACCESS ROADS ARE TO BE CONSTRUCTED TO HAVE A POSITIVE DRAINAGE FLOW AWAY FROM THE SITE AND EQUIPMENT. ANY DISCREPANCIES IN THE DRAWINGS OR SPECIFICATION MUST BE BROUGHT TO THE ATTENTION OF ENGINEER IMMEDIATELY.

7. ALL ACCESS ROAD AND SITE AREAS WILL HAVE AN UNDERLAYMENT OF MIRAFI-500X, OR EQUAL.

8. A SOIL STERILIZER SHALL BE APPLIED TO ALL GRAVEL SURFACES AND BE AN EPA REGISTERED LIQUID COMPOSITION AND OF PRE-EMERGENCE DESIGN THE PRODUCT LABEL AND INFORMATION WILL BE GIVEN TO ENGINEER/PROJECT MANAGE.

9. GRAVEL FILL SHALL BE WELL-GRADED, HARD, DURABLE, NATURAL SAND AND GRAVEL, FREE FROM ICE AND SNOW, ROOT, SOD RUBBISH AND OTHER DELETERIOUS OR ORGANIC MATTER. MATERIAL SHALL CONFORM TO THE FOLLOWING GRADUATION REQUIREMENTS.

SIEVE NUMBER	% PASSING BY WEIGHT
1/2"	100
#4	40-75
#10	30-60
#40	10-30
#100	5-15
#200	0-5

10. ALL FILL DIRT SHALL BE CLEAN AND NATURAL, FREE FROM ANY DELETERIOUS MATERIALS, ROOTS, ICE, SNOW, SOD AND RUBBISH.

11. ALL TRENCHES SHALL HAVE A SIX-INCH BASE OF CLEAN SAND FILL TO ACCEPT THE CONDUITS AND ANOTHER 12" OF CLEAN SAND FILL ON TOP OF THE CONDUITS. THE REMAINDER OF THE TRENCH SHALL HAVE A CLEAN COMPATIBLE FILL PLACED IN MAXIMUM LIFTS OF 8" AND MECHANICALLY COMPACTED TO A DENSITY OF 95%. METALLIC WARNING TAPE SHALL BE PLACED AT 12" BELOW FINISHED GRADE ALONG THE ENTIRE TRENCH.

12. ALL COMPACTION OF SITE AREAS SHALL BE ACCOMPLISHED BY MECHANICAL MEANS. LARGER AREAS SHALL BE COMPACTED BY A SHEEP'S FOOT VIBRATORY ROLLER WEIGHING AT LEAST 5 TONS. SMALLER AREAS SHALL BE COMPACTED BY A POWER-DRIVEN HAND HELD TAMPER. ALL COMPACTED AREAS SHALL BE COMPACTED TO WITHIN 95% AND TESTED BY AN INDEPENDENT LABORATORY. IT WILL BE THE RESPONSIBILITY OF THE CONTRACTOR TO SCHEDULE THE TESTING. ANY SCHEDULING FOR A PROCTOR WILL BE THE CONTRACTOR'S RESPONSIBILITY AND MUST BE DONE IN A TIMELY FASHION TO ASSURE THE PROJECT WILL NOT BE DELAYED.

13. ANY FILL OR STONE WILL BE PLACED IN MAXIMUM 8" LIFTS PRIOR TO COMPACTING.

14. FINISHED GRADE, INCLUDING TOP SURFACE COURSE, SHALL EXTEND A MINIMUM OF 12" BEYOND THE SITE FENCE AND SHALL COVER THE AREA AS INDICATED.

15. USE RIPRAP IN ANY AREAS WITH A SLOPE GREATER THAN 2:1 IN THE ENTIRE DITCH FOR 6' FEET IN ALL DIRECTIONS AT CULVERT OPENINGS AND WHEREVER INDICATED ON PLANS.

16. SEED, FERTILIZER AND STRAW COVER SHALL BE APPLIED TO ALL OTHER DISTURBED AREAS, DITCHES, DRAINAGE AND SWALES NOT OTHERWISE RIP RAPPED. SEED AND FERTILIZER SHALL BE APPLIED TO SURFACE CONDITIONS THAT WILL ENCOURAGE ROOTING. PREPARE SURFACE PROPERLY TO ACCEPT THE SEEDS. SOW SEEDS IN TWO OPPOSITE DIRECTIONS IN TWICE THE QUANTITY RECOMMENDED BY THE SEED PRODUCER.

17. THE CONTRACTOR IS RESPONSIBLE TO ENSURE GROWTH OF THE SEEDED AND LANDSCAPED AREAS BY WATERING, STRAW, MULCH, NET AND ANY OTHER PROPER LANDSCAPING METHOD NECESSARY. ALL AREAS MUST HAVE A SUSTAINED GROWTH BY THE TIME OF COMPLETION OF THE PROJECT.

18. THE CONTRACTOR SHALL PROTECT ALL AREAS FROM WASHOUTS AND SOIL EROSION. STRAW BALES SHALL BE PLACED AT THE INLET APPROACH TO ALL NEW OR EXISTING CULVERTS.

CONCRETE

1. ALL MATERIALS, LABOR AND METHODS SHALL CONFORM TO ALL APPLICABLE ASTM AND ACI STANDARDS AND REQUIREMENTS. IT IS THE CONTRACTOR'S RESPONSIBILITY TO ENSURE THAT ALL STANDARDS ARE MET. CONCRETE AND MATERIALS SHALL BE TESTED BY AN INDEPENDENT 3RD PARTY. THESE RESULTS SHOULD BE AVAILABLE UPON REQUEST.

2. THE CONTRACTOR SHALL SCHEDULE ALL INSPECTIONS AT THE TESTING LABORATORY AND INFORM ENGINEER WITHIN 24 HOURS OF INSPECTION OR TEST. THE TEST WILL INCLUDE SLUMP, AIR ENTRAINMENT, TEMPERATURE AND TEST CYLINDERS. A TOTAL OF FIVE CYLINDERS SHALL BE TAKEN DURING EACH POUR OR FOR EACH 50 YARDS IF THE POUR IS OVER 50 YARDS. THE FIRST CYLINDER WILL BE BROKEN AT 3-1/2 DAYS, A SECOND AT 7 DAYS, A THIRD AT 14 DAYS AND A FOURTH AT 28 DAYS. THE LAST CYLINDER WILL BE KEPT SEPARATELY FOR FUTURE USE IF NECESSARY.

3. ALL CONCRETE FOR THE PROJECT SHALL HAVE 28-DAY STRENGTH OF 4,000 PSI AND A SLUMP OF 5" MAXIMUM WITH THE MIXTURE PROPORTIONS MEETING THE ASTM AND ACI REQUIREMENTS. ALL ADMIXTURES AND MATERIALS USED FOR THE CONCRETE MIXTURE SHALL ALSO CONFORM TO THE PROPER ASTM REQUIREMENTS.

4. THE CONTRACTOR WILL INSTRUCT THE CONCRETE SUPPLIER TO HAVE AN APPROVED PLASTICIZER ON THE TRUCK TO AID IN THE POURING AND PLACING OF PIERS AND FOUNDATIONS.

5. AIR ENTRAINING AGENT SHALL BE USED IN ANY CONCRETE EXPOSED TO FREEZING TEMPERATURES.

6. THE CONTRACTOR IS RESPONSIBLE FOR ALL EMBEDMENT LOCATIONS, SIZES AND PROPER ANCHORING TO HOLD THE ITEMS IN PLACE ACCURATELY.

7. THE CONTRACTOR WILL SUPPLY AND USE A RELEASING AGENT ON ALL FORMS ANY ATTACHMENT TO EXISTING CONCRETE AREAS WILL BE DONE WITH SMOOTH DOWELS AT 18" O/C WITH A MINIMUM 6" EMBEDMENT IN EACH DIRECTION AND AN APPROVED BONDING AGENT TO BE APPLIED TO THE EXISTING SURFACE.

8. ALL EXPOSED CONCRETE WILL HAVE A UNIFORM FINISH AND BE SLOPED FOR DRAINAGE OFF THE SURFACE UNLESS OTHERWISE NOTED IN THE PLANS OR SPECIFICATIONS.

9. THE CONTRACTOR SHALL PROTECT THE CONCRETE FROM ANY ADVERSE WEATHER CONDITIONS; PREMATURE DRYING AND MECHANICAL INJURY TO INSURE PROPER CURING.

10. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE USE AND CONTROL OF ANY MECHANICAL EQUIPMENT TO INSURE THE PROPER PLACEMENT OF THE CONCRETE TO PREVENT ANY VOIDS OR GAPS.

METALS:

1. ALL LABOR, MATERIALS, FABRICATION AND WORKMANSHIP SHALL CONFORM TO THE CODES AND STANDARDS OF ASTM AND AWS.

2. ALL WORK SHALL BE DONE BY A CERTIFIED WELDER. THE CONTRACTOR SHALL SUBMIT A COPY OF THE CERTIFICATION TO OWNER.

3. ALL STEEL USED SHALL BE HOT DIPPED GALVANIZED.

WARRANTIES:

1. THE CONTRACTOR SHALL WARRANTY ALL GENERAL WORK FOR A MINIMUM OF 18 MONTHS OR AS STATED ELSEWHERE IN THE CONTRACT.

2. ROOF WARRANTY: ALL WORK SHALL BE PERFORMED IN FULL COMPLIANCE WITH THE ROOF MANUFACTURER AND CONTRACTOR SHALL OBTAIN WRITTEN VERIFICATION FROM ROOF MANUFACTURER THAT THE WARRANTY REMAINS VALID AFTER CONSTRUCTION IS COMPLETE.

ELECTRICAL

1. THE CONTRACTOR IS RESPONSIBLE FOR ALL PERMITTING, FILING AND FEES IN CONJUNCTION WITH THE PROJECT.

2. THE CONTRACTOR SHALL SCHEDULE ALL NECESSARY INSPECTIONS WITH THE PROPER AUTHORITIES AND INFORM ENGINEER 48-HOURS IN ADVANCE. ALL TICKETS AND INSPECTION VERIFICATIONS WILL BE SUBMITTED TO ENGINEER WITHIN 24 HOURS AFTER THE INSPECTION HAS TAKEN PLACE.

3. ALL LABOR AND MATERIAL SHALL CONFORM TO ALL LOCAL, STATE AND NATIONAL CODES, RULES, REGULATIONS AND STANDARDS.

4. ALL EQUIPMENT, WIRING AND MATERIALS MUST HAVE A UL LABEL.

5. ALL WORK SHALL BE DONE BY QUALIFIED AND EXPERIENCED JOURNEYMEN AND PERFORMED IN A WORKMAN LIKE MANNER AND SHALL PROCEED IN AN ORDERLY MANNER SO AS NOT TO HOLD UP THE PROGRESS OF THE PROJECT.

6. THOROUGHLY TEST ALL LINES, FEEDERS, EQUIPMENT AND DEVICES WITH MAXIMUM LOADS TO ASSURE PROPER OPERATION.

7. CONDUITS AND FITTINGS FOR OUTSIDE APPLICATIONS SHALL BE RIGID OR NON-METALLIC UNLESS OTHERWISE NOTED. CONDUITS AND FITTINGS FOR INSIDE BUILDING RUN ABOVE CEILINGS SHALL BE EMT. SUPPORT ALL CONDUITS WITHIN 3 FEET OF ALL JUNCTION BOXES AND AT 8'-FOOT INTERVALS OR LESS AS NEEDED WITH APPROVED ANCHORING DEVICES.

8. ALL WIRES SHALL BE COPPER; USE OF ALUMINUM CONDUCTORS WILL NOT BE PERMITTED. SEE ELECTRICAL PLANS FOR SIZING AND LOCATIONS. USE PROPER SIZE CONNECTORS PER LOCAL, STATE AND NATIONAL CODES.

9. CONDUCTOR LENGTHS SHALL BE CONTINUOUS FROM TERMINATION TO TERMINATION WITHOUT SPLICES.

10. PROVIDE PULL BOXES WHERE SHOWN AND WHERE REQUIRED BY CODE AND UTILITY COMPANIES.

11. ALL CONDUITS ROUGH IN SHALL BE COORDINATED WITH THE MECHANICAL EQUIPMENT TO AVOID LOCATIONS CONFLICTS. CONTRACTOR SHALL VERIFY ALL LOCATIONS.

12. ALL WIRES SHALL BE TAGGED AT ALL PULL BOXES, J-BOXES, EQUIPMENT BOXES AND CABINETS WITH APPROVED PLASTIC TAGS.

13. ALL BREAKERS IN CLOSED CENTER SHALL BE IDENTIFIED WITH TYPE WRITTEN LABELS NEATLY PLACED ALONG SIDE OF THE BREAKER.

14. ALL GROUND BUS BARS SHALL BE 1/4 "X 4" X 20" UNLESS OTHERWISE NOTED AND SHALL BE TINNED COPPER AND LARGE ENOUGH TO ACCOMMODATE THE REQUIRED NUMBER OF GROUNDING CONNECTIONS. THE HARDWARE SECURING ALL GROUND BARS SHALL BOND THEM TO THE STRUCTURE TO WHICH THEY ARE FASTENED.

15. ANY PENETRATIONS THROUGH WELLS, CEILINGS, FLOORS, ROOF OR ANY OTHER SURFACE SHALL BE CORE DRILLED AND CLOSED WITH A UL RATED FIRE ASSEMBLY. IF A FIRE RATING IS NOT REFERRED TO ON THE DRAWINGS, THEN THE CONTRACTOR SHALL ASSUME A 1-HOUR RATING.

16. THE MATERIALS AND EQUIPMENT INFORMATION ON THESE DRAWINGS IS BASED ON THE BEST AVAILABLE INFORMATION AT THE TIME THE DRAWINGS WERE ISSUED. CHANGES MAY OCCUR AND THE CONTRACTOR SHALL VERIFY ALL ITEMS PRIOR TO PLACEMENT AND MAKE ANY CORRECTIONS WITHOUT FURTHER COST TO OWNER.

17. ALL UNDERGROUND CONDUITS SHALL BE SCHEDULE 40 PVC. ALL SWEEPS OR BENDS AND ABOVE-GROUND CONDUITS SHALL BE RIGID GALVANIZED STEEL OR PVC SCHEDULE 80. ALL CONDUITS NOT TERMINATING INTO A CLOSED AREA MUST BE SEALED TO PREVENT ENTRY OF ANY MOISTURE OR FOREIGN OBJECTS. ALL CONDUIT RISERS TO INCLUDE SLIP TYPE EXPANSION JOINT.

GROUNDING:

1. THE CONTRACTOR SHALL VERIFY WITH OWNER THAT ALL CONDUIT SYSTEMS, EQUIPMENT, DEVICES, PANELS, LINES TRAYS, BRIDGES, PLATFORMS AND ANY/ALL ITEMS ARE GROUNDED PROPERLY.

2. THE GROUNDING SYSTEM CONNECTIONS SHALL BE MADE WITH EXOTHERMIC WELDS AND/OR MECHANICAL TWO-LUG COMPRESSION CONNECTORS AS INDICATED ON PLANS. USE ONLY STAINLESS STEEL SCREWS, BOLT, WASHERS AND NUTS FOR FASTENING.

3. CLEAN SURFACES THOROUGHLY BEFORE APPLYING GROUND LUGS OR CLAMPS. IF SURFACE IS COATED, THE COATING MUST BE REMOVED DOWN TO THE BARE METAL. AFTER THE COATING HAS BEEN REMOVED APPLY A NON-CORROSIVE COMPOUND TO THE CLEANED SURFACE AND INSTALL LUGS OR CLAMPS. WHERE GALVANIZING IS REMOVED FROM METAL, IT SHALL BE PAINTED OR TOUCHED UP WITH COLD GALVANIZING PAINT SUCH AS GLAVMOR OR EQUAL.

4. ALL CLAMPS OR SLEEVES AND SUPPORTS USED TO SUPPORT OR CHANNEL THE GROUNDING SYSTEM CONDUCTORS AND PVC CONDUITS SHALL BE PVC TYPE NON-CONDUCTIVE. DO NOT USE METAL BRACKETS OR SUPPORTS THAT WOULD FORM A COMPLETE RING AROUND ANY GROUNDING CONDUCTOR.

5. ALL GROUNDING CONNECTIONS SHALL BE COATED WITH T&B KOPR SHIELD ANTI-CORROSIVE AGENT. NO SUBSTITUTES ARE PERMITTED. VERIFY THE PRODUCE WITH OWNER PRIOR TO USAGE.

6. THE CONTRACTOR, UPON COMPLETION OF THE GROUNDING SYSTEM, SHALL SCHEDULE A TEST OF THE SYSTEM. ENGINEER REPRESENTATIVE WILL BE INFORMED 48 HOURS IN ADVANCE OF THE TEST.

PART 2 – PRODUCTS

2.1 GENERAL

A. GENERAL REQUIREMENTS: ALL MATERIALS AND EQUIPMENT SHALL BE IN ACCORDANCE WITH CONTRACT DOCUMENTS AND STANDARD PRODUCTS OF THE VARIOUS MANUFACTURERS, WITH ALL MATERIALS AND EQUIPMENT TO BE NEW, CLEAN, UNDAMAGED AND FREE OF DEFECTS AND CORROSION.

B. ACCEPTABLE PRODUCTS: THE PRODUCT OF A SPECIFIED OR APPROVED MANUFACTURER WILL BE ACCEPTABLE ONLY WHEN THE PRODUCT COMPLIES WITH OR IS MODIFIED AS NECESSARY TO COMPLY WITH ALL REQUIREMENTS OF THE CONTRACT DOCUMENTS.

C. COMMON ITEMS: WHERE MORE THAN ONE OF ANY SPECIFIC ITEM IS REQUIRED, ALL SHALL BE OF THE SAME TYPE AND MANUFACTURER.

D. UL LISTING: ALL MATERIALS AND EQUIPMENT SHALL BE UNDERWRITERS LABORATORIES (UL) LISTED AND LABELED, WHERE UL STANDARDS AND LISTINGS EXIST FOR SUCH MATERIALS OR EQUIPMENT.

2.2 RACEWAY SYSTEMS

A. RACEWAY SIZING: AS REQUIRED BY NEC (MINIMUM) WITH OVERSIZED RACEWAYS AS INDICATED ON DRAWINGS. MINIMUM CONDUIT SIZE: 3/4 – INCH UNLESS INDICATED OTHERWISE.

B. RACEWAY TYPES: RIGID GALVANIZED STEEL CONDUIT, ELECTRICAL METALLIC TUBING (EMT), FLEXIBLE STEEL CONDUIT, LIQUID-TIGHT FLEXIBLE STEEL CONDUIT, SCHEDULE 40 HEAVY WALL AND SCHEDULE 80 EXTRA HEAVY WALL RIGID NON-METALLIC (PVC) CONDUIT, CONFORMING TO APPLICABLE ANSI, NEMA AND UL STANDARDS.

C. FITTINGS: ALL RACEWAY FITTINGS (EXCEPT FOR RIGID NON-METALLIC CONDIT) TO BE STEEL OR MALLEABLE IRON AND UL LISTED FOR THE INTENDED APPLICATION. EMT FITTING TO BE COMPRESSION TYPE.

D. OUTLET BOXES (SURFACE MOUNTED): CADMIUM PLATED CAST OR MALLEABLE IRON.

E. PULL AND JUNCTION BOXES AND WIREWAYS: USE AS INDICATED AND REQUIRED. JUNCTION BOXES FOR GENERAL INDOOR USE (DRY LOCATIONS) TO BE OF GALVANIZED CODE GAUGE STEEL CONSTRUCTION MINIMUM 4" SQUARE X 1 ½ " DEEP WITH SCREW-ON COVERS. WIREWAYS TO BE UL LISTED, SHEET STEEL CONSTRUCTION WITH SCREW-ON COVERS. FOR EXTERIOR AND DAMP OR WET INDOOR LOCATIONS, USE BOXES AND WIREWAYS APPROVED FOR SUCH USE.

F. PIPE SLEEVES: RIGID STEEL CONDUIT OR IRON PIPE.

G. CONDUIT SEALS FOR CAST-IN-PLACE CONCRETE APPLICATIONS-ACCEPTABLE MANUFACTURERS: O-Z-/ GEDNEY TYPE "FSK"; THUNDER LINE CORPORATION "LINK SEAL" WITH "LINK SEAL" WALL SLEEVE. FOR CORE-DRILLED AND PRE-CAST OPENING APPLICATIONS - ACCEPTABLE MANUFACTURERS: O-Z/ GEDNEY TYPE "CSML", THUNDER LINE CORP. "LINK SEAL".

H. SWEEPS: ALL SWEEPS FOR COMMUNICATION EQUIPMENT OR INTO CONCRETE PAD SHALL BE MIN 24" RADIUS.

2.3 CONDUCTORS – 600 VOLT AND BELOW

A. GENERAL: SINGLE-CONDUCTOR 98% CONDUCTIVITY, ANNEALED, UNCOATED COPPER CONDUCTORS WITH 600-VOLT RATED TYPE "THHN/THWN" INSULATION.

B. CONNECTORS: NYLON SHELL INSULATE METALLIC SCREW-ON CONNECTORS FOR #14-#10 AWG AND BOLTED PRESSURE OR COMPRESSION TYPE LUGS AND CONNECTORS WITH INSULATING COVERS FOR #8 AWG AND LARGER.

2.4 HANGERS AND SUPPORTS

A. GENERAL: ALL HANGERS, SUPPORTS, FASTENERS AND HARDWARE SHALL BE HOT-DIPPED GALVANIZED OR EQUIVALENT CORROSION RESISTANCE BY TREATMENT AND SHALL BE MANUFACTURED PRODUCTS DESIGNED FOR THE APPLICATION.

TYPES: HANGERS, STRAPS, RISER SUPPORTS, CLAMPS, U-CHANNEL, THREADED RODS, ETC. AS INDICATED OR REQUIRED.

2.5 ELECTRICAL IDENTIFICATION

A. NAMEPLATES: THREE-LAYER LAMINATED PLASTIC WITH MINIMUM 3/16" HIGH WHITE ENGRAVED CHARACTERS ON BLACK BACKGROUND AND PUNCHED FOR MECHANICAL FASTENING.

B. FASTENERS: SELF-TAPPING STAINLESS-STEEL SCREWS OF NUMBER 10-32 STAINLESS STEEL MACHINE SCREWS WITH NUTS AND FLAT LOCK WASHERS.

C. UNDERGROUND WARNING TAPE: SIX-INCH WIDE POLYETHYLENE TAPE, PERMANENTLY BRIGHT COLORED WITH CONTINUOUS -PRINTED LEGEND INDICATING GENERAL TYPE OF UNDERGROUND LINE BELOW AND "CAUTION" COLORS AS FOLLOWS: 1) RED = ELECTRIC, 2) ORANGE = COMMUNICATIONS.

D. MARKING PENS: PERMANENT, WATERPROOF, QUIC

B. COOPERATION: COOPERATE WITH THE OTHER CONTRACTORS AND INDIVIDUAL DISCIPLINE FOR PLACEMENT, ANCHORAGE AND ACCOMPLISHMENT OF THE WORK RESOLVE INTERFERENCES BETWEEN WORK OF OTHER DISCIPLINES OR CONTRACTORS PRIOR TO COMMENCING INSTALLATION.

C. SUPPORTS AND SLEEVES: COORDINATE THE INSTALLATION OF REQUIRED SUPPORTING DEVICES AND SLEEVES TO BE SET IN POURED-IN-PLACE CONCRETE AND OTHER STRUCTURAL COMPONENTS AS THEY ARE CONSTRUCTED.

D. OBSTACLES AND INTERFERENCES: WHEN INSTALLING EQUIPMENT AND RACEWAYS, PROVIDE OFFSETS, FITTINGS, ACCESSORIES AND CHANGES IN ELEVATION OR LOCATION AS NECESSARY TO AVOID OBSTACLES AND INTERFERENCES, PER ACTUAL FIELD CONDITIONS.

3.4 EQUIPMENT PROTECTION

A. PROTECT ALL EQUIPMENT, MATERIALS AND WORK FROM THE WEATHER ELEMENTS, PAINT, MORTAR, CONSTRUCTION DEBRIS AND DAMAGE, UNTIL PROJECT IS SUBSTANTIALLY COMPLETE. REPAIR, REPLACE, CLEAN AND RETURN TO ORIGINAL CONDITION ALL DAMAGED AREAS, EQUIPMENT, UTILITIES, ETC. AND WORK SO AFFECTED.

3.5 LAYOUT

A. GENERAL: INSTALL MATERIALS AND EQUIPMENT LEVEL AND PLUMB AND PARALLEL AND PERPENDICULAR TO OTHER BUILDING SYSTEMS AND COMPONENTS, WHERE INSTALLED EXPOSED.

B. SERVICEABILITY: INSTALL EQUIPMENT AND RACEWAYS, ETC. TO READILY FACILITATE SERVICING, MAINTENANCE AND REPAIR OR REPLACEMENT OF COMPONENTS AND SO AS TO MINIMIZE INTERFERENCE WITH OTHER EQUIPMENT AND INSTALLATIONS.

C. CLEARANCES: PRIOR TO COMMENCING WORK, VERIFY THAT ALL EQUIPMENT WILL ADEQUATELY FIT AND CONFORM TO INDICATED AND CODE REQUIRED CLEARANCES, IN THE SPACES INDICATED ON THE DRAWINGS. IF REARRANGEMENT IS REQUIRED, SUBMIT PLAN AND ELEVATIONS DRAWINGS OR SKETCHES, INDICATING PROPOSED REARRANGEMENT, FOR THE ENGINEER'S APPROVAL. DO NOT REARRANGE WITHOUT EXPRESSED WRITTEN PERMISSION OF THE ENGINEER.

D. RIGHT-OF-WAY: WHEN LAYING OUT WORK, GIVE FIRST PRIORITY TO COMMUNICATIONS WIRING AND NEXT PRIORITY TO STEAM AND CONDENSATE LINES, SANITARY LINES, DRAIN LINES, FIRE PROTECTION PIPING AND SHEET METAL DUCT WORK. PROVIDE OFFSETS AS REQUIRED TO AVOID CONFLICTS. RESOLVE ALL CONFLICTS BEFORE COMMENCING INSTALLATION.

3.6 MOUNTING HEIGHTS

A. GENERAL: INDICATED HEIGHTS ARE MEASURED FROM THE CENTER OF THE DEVICE OUTLET BOX TO THE FINISHED FLOOR OR GRADE, UNLESS INDICATED OTHERWISE. REQUEST INSTRUCTIONS FOR MOUNTING HEIGHTS NOT INDICATED.

B. DEVICES: SWITCHES – 48" A.F.F. / RECEPTACLES – 18" A.F.F. / THERMOSTAT – 48" A.F.F.

3.7 HOLES, SLEEVES AND OPENINGS

A. GENERAL: PROVIDE ALL HOLES, SLEEVES AND OPENINGS REQUIRED FOR THE COMPLETIONS OF WORK, AND RESTORE ALL DAMAGED SURFACES TO MATCH SURROUNDING SURFACES. MAINTAIN INTEGRITY OF ALL FIRE AND SMOKE RATED BARRIERS USING APPROVED FIRESTOPPING SYSTEMS. WHEN CUTTING HOLES OR OPENINGS OR INSTALLING SLEEVES, DO NOT CUT, DAMAGE OR DISTURB STRUCTURAL ELEMENTS OR REINFORCING STEEL UNLESS APPROVED IN WRITING BY THE PROJECT STRUCTURAL ENGINEER.

B. CONDUIT PENETRATIONS: SIZE CORE DRILLED HOLES SO THAT AN ANNULAR SPACE OF NOT LESS THAN ¼" AND NOT MORE THAN ½" IS LEFT AROUND THE CONDUIT. WHEN OPENINGS ARE CUT IN LIEU OF CORE DRILLED, PROVIDE SLEEVE IN ROUGH OPENING. SIZE SLEEVES TO PROVIDE AN ANNULAR SPACE OF NOT LESS THAN ¼" AND NOT MORE THAN 1" AROUND THE CONDUIT, PIPE, ETC. PATCH AROUND SLEEVE TO MATCH SURROUNDING SURFACES.

3.8 RACEWAY SYSTEMS

A. UNLESS INDICATED OTHERWISE, USE RACEWAY TYPES AS FOLLOWS:

1) OUTDOORS EXPOSED: RIGID GALVANIZED STEEL CONDUIT AND SUNLIGHT RESISTANT SCHEDULE 40, WHERE NOT SUBJECT TO PHYSICAL DAMAGE.

2) LIQUID TIGHT FLEXIBLE STEEL CONDUIT: USE WHERE FLEXILE STEEL CONDUIT CONNECTIONS ARE REQUIRED IN DRY, DAMP, WET OR OILY LOCATIONS, FOR CONNECTION TO TRANSFORMERS, VIBRATING EQUIPMENT, EQUIPMENT REQUIRING MINOR ADJUSTMENTS IN POSITIONS AND FOR FINAL CONNECTIONS TO ALL MOTORS AND SIMILAR EQUIPMENT.

B. RACEWAY ROUTING: AS REQUIRED BY JOB CONDITIONS UNLESS SPECIFIC ROUTES OR DIMENSIONED POSITIONS ARE INDICATED ON THE DRAWINGS, INSTALL TIGHT TO SLAB, BEAMS AND JOISTS WHEREVER POSSIBLE. ROUTE EXPOSED CONDUIT AND CONDUIT INSTALLED ABOVE CEILINGS, PARALLEL OR PERPENDICULAR TO WALLS, CEILINGS AND STRUCTURAL MEMBERS. INSTALL TO MAINTAIN MINIMUM HEADROOM AND TO PRESENT A NEAT APPEARANCE. RUN PARALLEL RACEWAYS TOGETHER WITH BENDS MADE FROM THE SAME CENTERLINE. VERIFY EXACT LOCATIONS OF ALL RACEWAYS, PULL BOXES AND JUNCTION BOXES. RESOLVE ANY CONFLICTS BEFORE INSTALLATION. DO NOT INSTALL HORIZONTAL CONDUIT RUN BELOW 7' – 6" AFF IN EXPOSED LOCATIONS.

1) DO NOT ROUTE ANY CONDUITS INTO OR THRU, STAIRWAYS, ELEVATORS SHAFTS OR ELEVATOR MACHINE ROOM.

2) CONCEAL CONDUITS ABOVE, SUSPENDED CEILINGS WHERE POSSIBLE.

3) COORDINATE ALL CONDUIT STUB UPS WITH ROOF TOP EQUIPMENT LAYOUT.

4) COORDINATE INSTALLATIONS OF CONDUITS, EQUIPMENT, ETC. WITH BUILDING OWNER PRIOR TO COMMENCEMENT.

C. RACEWAY INSTALLATION: CUT CONDUIT ENDS SQUARE USING SAW OR PIPECUTTER AND REAM EACH CUT END SMOOTH. CAREFULLY MAKE ALL CONDUIT BENDS AND OFFSETS SO THAT THE INSIDE DIAMETER OF PIPE IS NOT REDUCED. MAKE BENDS SO THAT LEGS ARE IN THE SAME PLANE. MAKE OFFSETS SO THAT LEGS ARE IN THE SAME PLANE AND PARALLEL. PROTECT STUB-UPS FROM DAMAGE AND CAREFULLY REBEND WHEN NECESSARY.

D. FITTINGS: MAKE UP ALL RACEWAY FITTINGS TIGHT SO THAT FINAL INSTALLATION OF RACEWAY, FITTINGS AND ENCLOSURES CONSTITUTES A FIRM MECHANICAL ASSEMBLY AND A CONTINUOUS ELECTRICAL CONTINUITY. PROVIDE INSULATING BUSHINGS ON CONDUIT TERMINATIONS.

E. PROTECTION: PROTECT ALL RACEWAYS, ENCLOSURES AND EQUIPMENT DURING CONSTRUCTION TO PREVENT ENTRY OF CONCRETE, DEBRIS AND OTHER FOREIGN MATTER. FREE CLOGGED CONDUITS OF ALL OBSTRUCTIONS OR REPLACE PRIOR TO PULLING WIRE. DO NOT PULL WIRE WITHIN BUILDINGS UNTIL BUILDINGS ARE COMPLETELY ENCLOSED.

F. BOXES: INSTALL ALL OUTLET PULL AND JUNCTION BOXES RIGIDLY, PLUMB AND LEVEL, SUPPORT AND SECURE BOXES INDEPENDENTLY FROM CONDUITS TERMINATING AT BOX. INSTALL ALL BOXES SO AS TO BE ACCESSIBLE AND SO THAT COVERS MAY BE EASILY REMOVED.

G. PROVIDE CAP TO PREVENT ENTRY OF MATERIALS AND MOISTURE FOR ALL SPARE CONDUITS.

H. CONDUIT SEALS: INSTALL CONDUIT SEAL FOR EACH CONDUIT PENETRATING AT AN EXTERIOR BUILDING WALL, BELOW GRADE (UNLESS PENETRATION IS BELOW LOWEST BUILDING FLOOR SLAB) AND ELSEWHERE AS INDICATED AND SO AS TO ACHIEVE A SEALED WATERTIGHT INSTALLATION.

I. RIGID GALVANIZED STEEL CONDUIT SHALL BE USED IN AREAS SUBJECT TO PHYSICAL DAMAGE. CUT ENDS WILL BE REAMED, THREADED AND COLD GALVANIZED (COMPRESSION FITTINGS WILL NOT BE ACCEPTED).

3.9 CONDUCTORS – 600 VOLT AND BELOW

A. MINIMUM CONDUCTOR SIZE: ALL BRANCH CIRCUIT WIRING SHALL BE MINIMUM #12 AWG. FEEDER CIRCUIT WIRING SHALL BE #2 AWG UNLESS OTHERWISE NOTED. ALL CONTROL CIRCUIT WIRING SHALL BE MINIMUM #14 AWG, UNLESS INDICATED OTHER WISE. PROVIDE LARGER SIZES, AS INDICATED OR REQUIRED. A FULL SIZED EQUIPMENT GROUNDING CONDUCTOR WILL BE INSTALLED IN POWER AND LIGHTING CONDUITS. (CONDUITS WILL NOT BE USED AS A GROUNDING CONDUCTOR)

B. IN RACEWAY: INSTALL ALL WIRING IN CONDUIT OR OTHER SPECIFIED RACEWAY, UNLESS INDICATED OTHERWISE. ROMEX, BX, AC AND MC TYPE CABLES ARE NOT ALLOWED.

C. TERMINATIONS: FURNISH AND INSTALL TERMINATIONS INCLUDING LUGS IF NECESSARY TO MAKE ALL ELECTRICAL CONNECTIONS INDICATED OR REQUIRED. EXTERIOR AND SWITCHGEAR TERMINATIONS SHALL HAVE AN ANTI-OXIDANT APPLIED. SPRING TYPE WIRE CONNECTORS USED IN EXTERIOR BOXES SHALL BE SILICONE FILLED.

3.10 HANGERS AND SUPPORTS

A. GENERAL: RIGIDLY SUPPORT AND SECURE ALL MATERIALS, RACEWAY EQUIPMENT TO BUILDINGS STRUCTURE USING HANGERS, SUPPORTS AND FASTENER SUITABLE FOR THE USE, MATERIALS AND LOADS ENCOUNTERED. PROVIDE ALL NECESSARY HARDWARE. PROVIDE CONDUIT SUPPORTS AT MAXIMUM 5 FEET ON CENTER.

B. OVERHEAD MOUNTING: ATTACH OVERHEAD MOUNTED EQUIPMENT TO STRUCTURAL FRAMEWORK OR SUPPORTING METAL FRAMEWORK. DO NOT MAKE ATTACHMENTS TO STEEL ROOFING, STEEL FLOORING OR CEILING MINERAL TILE.

C. WALL MOUNTING: SUPPORT WALL MOUNTED EQUIPMENT BY MASONRY, CONCRETE BLOCK, METAL FRAMING OR SUB-FRAMING.

D. EXTERIOR WALLS: MOUNT ALL EQUIPMENT LOCATED ON THE INTERIOR OR EXTERIOR BUILDING WALLS, AT LEAST ONE INCH AWAY FROM WALL SURFACE, USING SUITABLE SPACERS.

E. STRUCTURAL MEMBERS: DO NOT CUT, DRILL OR WELD ANY STRUCTURAL MEMBER EXCEPT AS SPECIFICALLY APPROVED BY THE ENGINEER.

F. INDEPENDENT SUPPORT: DO NOT SUPPORT MATERIALS OR EQUIPMENT FROM OTHER EQUIPMENT, PIPING, DUCTWORK OR SUPPORT FOR SAME.

G. TEMPORARY CONDITIONS: DO NOT ATTACH OR SUPPORT ELECTRICAL WORK FROM REMOVABLE OR KNOCKOUT PANELS OR TEMPORARY WALLS OR PARTITIONS

H. RACEWAY SUPPORTS: RIGIDLY SUPPORT ALL RACEWAY WITH MAXIMUM SPACING'S PER NEC AND SO AS TO PREVENT DISTORTION OF ALIGNMENT DURING PULLING OPERATION. USE APPROVED HANGERS, CLAMPS AND STRAPS FOR INDIVIDUAL RUNS. DO NOT USE PERFORATED STRAPS OR TIE WIRES. WHERE MULTIPLE PARALLEL RACEWAYS ARE RUN TOGETHER, USE TRAPEZE TYPE HANGER ARRANGEMENT MADE FROM U-CHANNEL AND ACCESSORIES, SUSPENDED BY THREADED RODS AND ALLOW AT LEAST 25% SPARE CAPACITY FOR FUTURE INSTALLATION OF ADDITIONAL RACEWAYS. RIGIDLY ANCHOR VERTICAL CONDUITS SERVING FLOOR-MOUNTED OR ISLAND TYPE EQUIPMENT MOUNTED AWAY FROM WALLS WITH METAL BRACKETS OR RIGID STEEL CONDUIT EXTENSION SECURED TO FLOOR.

I. MISCELLANEOUS SUPPORTS: PROVIDE ANY ADDITIONAL STRUCTURAL SUPPORT STEEL, BRACKETS, ANGLES, FASTENERS AND HARDWARE AS REQUIRED TO ADEQUATELY SUPPORTING ALL ELECTRICAL MATERIALS AND EQUIPMENT.

J. ONE HOLE STRAPS SHALL NOT BE USED FOR CONDUITS LARGER THAN ? INCH.

3.11 EQUIPMENT CONNECTIONS

A. VERIFICATION: OBTAIN AND REVIEW SHOP DRAWINGS, PRODUCT DATA AND MANUFACTURER'S INSTRUCTION FOR EQUIPMENT FURNISHED BY OTHERS. EXAMINE ACTUAL EQUIPMENT TO VERIFY PROPER CONNECTION LOCATIONS AND REQUIREMENTS.

B. ROUGH IN: PROVIDE ALL REQUIRED CONDUIT, BOXES, FITTINGS, WIRE, CONNECTORS AND MISCELLANEOUS ACCESSORIES, ETC. AS NECESSARY TO ROUGH IN THE MAKE FINAL CONNECTIONS TO ALL EQUIPMENT REQUIRING ELECTRICAL CONNECTIONS. IN GENERAL, MOTORS AND EQUIPMENT SHALL BE WIRED IN CONDUIT TO A JUNCTION BOX (OR SAFETY SWITCH) NEAR THE UNIT AND FROM THERE TO THE UNIT IN FLEXIBLE METAL OR LIQUID TIGHT FLEXIBLE STEEL CONDUIT.

C. CONNECTIONS: PROVIDE PROPERLY SIZED OVERLOAD AND SHORT CIRCUIT PROTECTION FOR ALL EQUIPMENT CONNECTED, WHETHER FURNISHED UNDER THIS CONTRACT OR BY OTHERS. VERIFY PRIOR CONNECTIONS WITH MANUFACTURER'S PUBLISHED DIAGRAMS AND COMPLY WITH SAME. VERIFY THAT EQUIPMENT IS READY FOR ELECTRICAL CONNECTIONS, WIRING AND ENERGIZATION, PRIOR TO PERFORMING SAME.

D. CONTROL WIRING: PROVIDE ALL CONTROL WIRING TO REMOTE DEVICES OR EQUIPMENT, AS INDICATED OR REQUIRED. MODIFY EQUIPMENT CONTROL WIRING, INSTALL OR DISCONNECT JUMPERS, ETC., AS REQUIRED.

3.12 IDENTIFICATION:

A. GENERAL: LOCATE NAMEPLATE MARKING OR OTHER IDENTIFICATION MEANS ON OUTSIDE OF EQUIPMENT OR BOX FRONT COVERS WHEN ABOVE CEILINGS AND WHEN IN MECHANICAL OR ELECTRICAL EQUIPMENT ROOMS OR OTHER UNFINISHED AREAS, AND ON INSIDE OF FRONT COVER WHEN N FINISHED ROOMS/AREAS. USE CONTRACT DOCUMENT DESIGNATIONS FOR IDENTIFICATION UNLESS INDICATED OTHERWISE.

B. NAMEPLATES: PROVIDE NAMEPLATE ENGRAVED WITH EQUIPMENT DESIGNATION FOR EACH SAFETY SWITCH, PANELBOARD, TRANSFORMER, MOTOR STARTER AND ALL OTHER EQUIPMENT, ETC.

C. UNDERGROUND WARNING TAPE: DURING TRENCH BACKFILLING FOR EACH UNDER GROUND ELECTRICAL, TELEPHONE, SIGNAL AND COMMUNICATIONS LINE, PROVIDE A CONTINUOUS UNDERGROUND WARNING TAPE LOCATED DIRECTLY ABOVE LINE, AT SIX TO EIGHT INCHES BELOW FINISHED GRADE.

D. MARKING PEN LABELING: MARK EACH JUNCTION AND PULL BOX INDICATING SOURCE DESIGNATION AND CIRCUIT NUMBER(S) FOR THE ENCLOSED CONDUCTORS.

E. LABEL ALL WIRES AND CABLES AT EVERY POINT OF TERMINATION AND IN ALL PULL BOXES AND JUNCTION BOXES. FOR POWER CIRCUITS, APPLY WIRE TAGS INDICATING APPROPRIATE CIRCUIT OR FEEDER NUMBER TO EACH CONDUCTOR PRESENT IN DISTRIBUTION PANEL AND PANELBOARD GUTTERS AND TO EACH CONDUCTOR IN PULL AND JUNCTION BOXES.

F. LOAD CENTER CIRCUIT DIRECTORIES: AT COMPLETION OF PROJECT, ACCURATELY COMPLETE EACH PANELBOARD CIRCUIT DIRECTORY CARD, IDENTIFYING LOAD SERVED OR "SPARE" OR "SPACE" FOR EACH CIRCUIT POLE. WHEN MODIFYING, ADDING OR DELETING CIRCUITS AT AN EXISTING PANELBOARD, UPDATE THE EXISTING (OR PROVIDE NEW) CIRCUIT DIRECTORY CARD TO ACCURATELY REFLECT FINAL CONDITIONS.

3.13 GROUNDING SYSTEM

A. EXAMINATION: VERIFY THAT SURFACES ARE READY TO RECEIVE WORK AND THAT THE FIELD MEASUREMENTS ARE AS SHOWN ON SHOP DRAWINGS.

B. INSTALLATION – INSTALL AS INDICATED ON THE DRAWINGS AND AS REQUIRED:

1) GROUNDING NETWORK SYSTEM SHALL INCLUDE ELECTRICAL TRENCHING, BACKFILL AND COMPACTION PER THIS SECTION; GROUNDING WIRE, GROUNDING ELECTRODES AND ALL NECESSARY MATERIALS AND LABOR REQUIRED TO COMPLETE THE SPECIFIED SYSTEM.

2) ALL GROUND CONNECTIONS SHALL CONFORM TO IEE STD 837 AND/OR UL467 PROTECTED FROM CORROSION AS NOTED ON DWGS. USE CADWELD OR DOUBLE LUG FOR OUTDOORS ABOVE AND BELOW GRADE INDOORS AND ON ROOF INSTALLATIONS, AS INDICATED ON PLANS.

3) ALL CLAM CONNECTIONS SHALL BE MADE ACCESSIBLE FOR INSPECTION. USE ONLY WHERE SPECIFIED.

4) ALL UNDERGROUND CONNECTIONS SHALL BE INSPECTED BY OWNER REPRESENTATIVE PRIOR TO BACKFILLING.

5) IF ROCK IS ENCOUNTERED, GROUND ROD SHALL BE INSTALLED IN A 2 INCH DIAMETER DRILLED HOLE TO THE REQUIRED DEPTH AND BACKFILLED WITH APPLICATIONS OF MAGNESIUM SULFATE OR COPPER SULFATE.

6) GROUNDING WIRE SHALL NOT BEND LESS THAN 90° IN ANY LOCATION WITH A MINIMUM BEND RADIUS OF 8".

7) PROVIDE CORROSION PROOFING PAINT ON SURFACES EXPOSED DURING INSTALLATION.

8) ALL BELOW GROUND CONNECTIONS SHALL BE CADWELD. PROVIDE CORROSION PROTECTIONS TO CONNECTION AREA, IF INDICATED.

9) ALL ABOVE GROUND CONNECTIONS SHALL BE BOLTED CLAMP. BURNDY CONNECTIONS SHALL NOT BE USED FOR EXTERIOR GROUNDING, UNLESS NOTED.

10) ALL GROUND CONNECTIONS TO THE GROUND BAR SHALL BE MADE WITH DOUBLE HOLE HYDRAULICALLY INDENTED LUGS, EXCEPT FOR TWO CADWELDS, AS SHOWN ON PLANS.

11) WHERE GROUND CONNECTIONS ARE MADE, THE CONTACT POINTS ARE TO BE CLEANED AND MADE FREE OF FOREIGN MATERIALS, SUCH AS PAINT AND CORROSION, TO ENSURE AN ADEQUATE BOND.

12) ALL GROUNDING AND BONDING INSTALLATIONS AND CONNECTIONS SHALL COMPLY WITH OWNER WIRELESS SERVICES ELECTRICAL AND GROUNDING SPECIFICATIONS.

13) ALL MECHANICAL GROUNDING CONNECTIONS MADE THROUGHOUT THESE DRAWINGS SHALL BE MADE WITH KOPR-SHIELD. THERE IS NO EQUIVALENT FOR THESE ANTIOXIDATION COMPOUNDS. NO OTHER COMPOUND WILL BE ACCEPTED. COAT ALL WIRES BEFORE LUGGING. COAT ALL SURFACES BEFORE CONNECTING.

C. FIELD QUALITY CONTROL – FIELD INSPECTION AND TESTING WILL BE PERFORMED AS REQUIRED:

1) THE CONTRACTOR SHALL VERIFY THAT THE SYSTEM IS EFFECTIVELY GROUNDED, MEETS NEC ARTICLE 250 REQUIREMENTS AND IS ACCEPTABLE TO THE LOCAL UTILITY AND LOCAL AUTHORITY HAVING JURISDICTION.

2) THE CONTRACTOR SHALL MAKE ALL MEASUREMENTS REQUIRED TO TEST GROUNDING NETWORK SYSTEM. SYSTEM SHALL BE TESTED AT THE TIME OF INSTALLATION.

3) THE GROUNDING NETWORK SYSTEM TEST PROCEDURE SHALL COMPLY WITH NFPA 78 THREE POINT TECHNIQUE FOR GROUND RESISTANCE, EXCEPT AS MODIFIED HEREIN.

4) CONTRACTOR SHALL VERIFY THE ADEQUACY OF THE INSTALLED SYSTEM CONTRACTOR SHALL CONDUCT "SITE RESISTANCE TO EARTH GROUNDING TESTING" PER OWNER STANDARD. INSTALLED SYSTEM SHALL ACHIEVE A GROUND RESISTANCE OF LESS THAN 5 OHMS.

5) RECORD GROUND RESISTANCE TEST RESULTS ON OWNER "GROUND RESISTANCE TEST" FORM.

6) TEST SHALL BE WITNESSED BY OWNER REPRESENTATIVE.

7) COPIES OF THE TEST RESULTS SHALL BE SUBMITTED TO OWNER' REPRESENTATIVE & ENGINEER.

3.14 CHECKOUT, TESTING AND ADJUSTING

A. CORRECTION/REPLACEMENT: AFTER TESTING BY CONTRACTOR, OWNER OR ENGINEER CORRECTS ANY DEFICIENCIES AND REPLACES MATERIALS AND EQUIPMENT SHOWN TO BE DEFECTIVE OR UNABLE TO PERFORM AT DESIGN OR RATED CAPACITY.

B. POWER CONDUCTORS: CONTRACTOR SHALL CONDUCT CONTINUITY & INSULATION TEST ON CONDUCTOR'S BETWEEN SERVICE DISCONNECT SWITCH & POWER CABINET.

C. GROUNDING – IEEE FALL OF POTENTIAL TESTS: CONDUCT TEST WITH AN AEMC MODEL #4500 TESTER. THE METHOD OF USING TWO AUXILIARY GROUND RODS (AS DESCRIBED IN I.E.E.E STANDARD #81 – 1983, PART 1) SHALL BE USED. THE AUXILIARY TEST RODS MUST BE SUFFICIENTLY FAR AWAY FROM THE RODS SO THAT THE REGIONS IN WHICH THEIR RESISTANCE IS LOCALIZED DO NOT OVERLAP.

3.15 SYSTEMS DEMONSTRATION

A. INSTRUCT OWNER' REPRESENTATIVE(S) IN THE START-UP, OPERATION AND MAINTENANCE OF ALL ELECTRICAL SYSTEMS AND EQUIPMENT AS REQUESTED BY GENERAL DYNAMICS' REPRESENTATIVE.

3.16 CLEANING AND TOUCH-UP PAINTING

A. GENERAL: PERIODICALLY REMOVE FROM THE PROJECT SITE, ALL WASTE, RUBBISH AND CONSTRUCTION DEBRIS ACCUMULATED FROM CONSTRUCTION OPERATIONS. THE PREMISES SHALL BE LEFT CLEAN AND FREE OF ANY DEBRIS AND UNUSED CONSTRUCTION MATERIALS PRIOR TO FINAL ACCEPTANCE.

B. ELECTRICAL EQUIPMENT: REMOVE ALL DUST, DIRT, DEBRIS, MORTAR, WIRE SCRAPS, RUST AND OTHER FOREIGN MATERIALS FROM THE INTERIOR AND EXTERIOR OF ALL ELECTRICAL EQUIPMENT AND ENCLOSURES, AND WIPE DOWN.

C. TOUCH-UP PAINTING: RESTORE AND REFINISH TO ORIGINAL CONDITION ALL SURFACES OF ELECTRICAL EQUIPMENT THAT IS SCRATCHED, MARRED AND/OR DENTED DURING SHIPPING, HANDLING OR INSTALLATION. REMOVE ALL RUST AND PRIME AND PAINT AS RECOMMENDED BY THE MANUFACTURER.

PART 4 – CONTRACTOR DELIVERABLES

1. THE CONTRACTOR SHALL CARRY ON HIS WORK IN ACCORDANCE WITH ALL GOVERNING NATIONAL, STATE, COUNTY AND LOCAL CODES, AS WELL AS, OSHA REGULATIONS.

2. TESTING OF CAST-IN-PLACE CONCRETE, STRUCTURAL STEEL AND ANY OTHER MATERIAL USED SHALL BE AS REQUIRED BY AND IN ACCORDANCE WITH LOCAL REGULATIONS.

3. THE GROUNDING SYSTEM FOR EACH SITE SHALL BE TESTED FOR A GROUND RESISTANCE OF 5 OHMS. ON OF THREE TESTS SHALL BE USED: IEEE FALL-OF-POTENTIAL TEST, AN APPROVED CLAMP ON TEST OR A DIRECT METHOD (TWO-TERMINAL) "CONTINUITY" TEST. THE FALL-OF-POTENTIAL TEST IS THE PREFERRED TESTING METHOD AND SHALL BE USED WHENEVER POSSIBLE. ANY GROUND RESISTANCE OF GREATER THAN 5 OHMS BUT LESS THAN 25 OHMS SHALL BE REVIEWED BY OWNER AND AGREED UPON WITH AWS ON A CASE-BY-CASE BASIS.

4. AS-BUILT INFORMATION: TWO (2) COPIES OF A REPORT CONTAINING A DESCRIPTION OF THE TOWER/MONOPOLE, SKETCHES AND RED-LINED CONSTRUCTION DRAWINGS SHALL BE SUBMITTED. THE MAXIMUM SIZE OF ANY DRAWING IN THE REPORT SHALL BE "11X17". THE COVER SHEET OF THE REPORT SHALL CONTAIN THE SITE NAME AND SITE NUMBER. ALSO INCLUDED IN THIS REPORTS SHALL BE RED-LINED RF INFORMATION.

5. CONTRACTOR SHALL HAVE 72 HOURS TO COMPLETE PUNCH-LIST ITEMS FOUND INCOMPLETE OR INADEQUATE BY OWNER DURING SITE COMPLETION WALK.

6. ONE ORIGINAL OF EACH OF THE FOLLOWING SHALL BE PROVIDED TO OWNER UPON COMPLETION OF PROJECT:
   - CERTIFICATE OF AUTHENTICITY
   - ALL INSPECTION SIGN-OFFS
   - BUILDING AND ELECTRICAL PERMITS
   - RELEASE OF WAIVER AND LIENS
   - ANTENNA PACKETS (EQUIPMENT INFORMATION, PAPERS, ETC.
   - WARRANTIES ON ALL ITEMS INSTALLED AND WORK PERFORMED
   - PHOTOGRAPHS OF THE SITE BEFORE, DURING AND AFTER CONSTRUCTION
   - ALL THIRD- PARTY INSPECTION REPORTS (CONCRETE, PLATFORM, ETC.)
   - COMPLETE PUNCH LIST

7. ALL PERMITS ARE TO BE ON-SITE AT ALL TIMES, INCLUDING DURING AND AFTER CONSTRUCTION.

REVIEWED
MAY 16 2007
BOULDER COUNTY
BUILDING DIVISION

at&t

PEI

PORCELLO ENGINEERING INC.
6916 NORTH BROADWAY
DENVER, CO 80221
Phone 303.412.8996 Fax 303.412.8998
Consulting Engineers



Cascadia PM
5950 South Willow Drive, Suite 304
Greenwood Village, CO 80111
P: 303.694.7600 F: 303.600.7797

REV	DATE	DESCRIPTION	DRAWN BY	CHECKED BY	APP'D BY
+	5/08/07	CD PERMIT APPLICATION			

SITE NO.
1469

SITE NAME
LYONS

SITE ADDRESS
5145 UTE HWY
LYONS, CO

SHEET TITLE
GENERAL NOTES

SHEET NO.
GN-02
SHEET 11 of 11

Paid Date

PAID

23 MAY 2007

BOULDER COUNTY BUILDING DIV.

Boulder County

BUILDING PERMIT

Land Use Department ° Building Safety & Inspection Division

Courthouse Annex ° 2045 13th St ° 13th & Spruce Streets

P.O. Box 471 ° Boulder, Colorado 80306-0471

(303) 441-3925 Fax 303.441.4856

Permit Number

BP - 07 - 598

Inspector Copy

PROJECT LOCATION

Project Address	Unit Number	City	Zip Code
5134 UTE HW		LY	80540

Lot	Block	Subdivision Name	Section	Township	Range
		UNPLATTED	28	3N	70

WORK DESCRIPTION

Description of Work

INSTALL 3 GROUPS OF PANEL ANTENNAS

EXPIRED

OWNER

Owner Name & Address

CEMEX INC
5134 UTE HW

City	State	Zip Code
LONGMONT	CO	80503

Day / Primary Phone	Night / Secondary Phone
(303)-823-2100	

CONTRACTOR / AGENT

Contractor Name & Address

CASCADIA PM LLC
5050 S WILLOW DR #304

City	State	Zip Code
GREENWOOD VILLAGE	CO	80111

Day / Primary Phone	Night / Secondary Phone
(970)-227-6899	

STRUCTURE DATA

Activity Code	Description Code
04 Electrical	17 Non Structures(pasture, pond, road, overlot grading, subdiv)

Occupancy Group	Construction Type	Stories	Structure Height (FT)
U	VB		

Snow Load (PSF)	Wind Speed (MPH)	New Bedrooms	New Bathrooms

SETBACKS / ZONING / FEES

Setback #1 (FT)	Setback #2 (FT)	Setback #3 (FT)	Setback #4 (FT)	Property Size (Acres)
				866

Zoning District(s)	Parcel ID Number
GI	120328000001

Tax ID Number	Deed Number	Access Permit Number
53049	1756839	

New Construction Square Footage

	Finished	Unfinished
Basement:	0	0
1st Story:	0	0
2nd Story:	0	0
3rd Story:	0	0
All Other Stories:	0	0

Decks	Covered Area	Greenhouse	Garage

Fee Summary

Project Valuation:	22,000
Building Permit Fee:	353.33
Plan Check Fee:	230.00
Access Fee:	.00
Taxes:	71.50
Total Fees:	**654.83**

Receipt Number: 52635

Utilities & Services

Water Provider	Sanitation Provider
Not Known	Not Known
Electric Provider	**Gas Provider**
Poudre Valley REA	Not Known
Fire District	
Hygiene	

Approved By Chief Building Official	Date Of Issue
Jeff Dwight	May 23, 2007

The approval of plans and specifications does not permit the violation of any section of the Boulder County Building Code, Land Use Code, or any other Ordinance or State Law.

° FIELD INSPECT TO CODE °

Permit shall become null and void if construction is not commenced within 180 days of issuance, or if work is suspended or abandoned for a period of 180 days after the work is commenced.

5134 UTE HW.

5134 UTE HW.

REQUIREMENTS & REMARKS

Land Use Department ° Building Safety & Inspection Division
Courthouse Annex ° 2045 13th St ° 13th & Spruce Streets
P.O. Box 471 ° Boulder, Colorado 80306-0471
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Permit Number
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Inspector Copy

Building Code • Land Use Dept / Building Division / 303 - 441 - 3925

Comments

All work per the applicable provisions of the Boulder County Building Code to include: the 2003 IBC, 2003 IMC, 2003 IPC, 2003 IFGC, 2003 IECC, 2002 NEC, and the Boulder County Amendments.
New or replacement exterior lighting fixtures must be down-lit shielded fixtures in conformance with Article 7-1600 of the Boulder County Land Use Code.

5134 UTE HW.

5134 UTE HW.

BUILDING INSPECTION RECORD

Land Use Department ° Building Safety & Inspection Division
Courthouse Annex ° 2045 13th St ° 13th & Spruce Streets
P.O. Box 471 ° Boulder, Colorado 80306-0471
(303) 441-3925 Fax 303.441.4856

Permit Number
BP - 07 - 598
Site Address
5134 UTE HW

Inspector must date and initial each inspection pertaining to this job.
DO NOT POUR CONCRETE OR COVER ANY WORK UNTIL APPROVED BY INSPECTOR

ROUGH BUILDING INSPECTIONS

INSPECTION	DATE APPROVED	INSPECTOR	COMMENTS	INSPECTOR	DATE
FOOTINGS / PADS					
FOUNDATION WALLS & REINFORCING					
DAMP-PROOFING (BASEMENTS ONLY)					
UNDERGROUND ELECTRIC					
UNDERGROUND PLUMBING & PRESSURE TEST					
UNDERGROUND GAS PIPING & PRESSURE TEST					
ROUGH ELECTRIC	12/24/07		COVER	12/24/07	
ROUGH PLUMBING & PRESSURE TEST					
ROUGH GAS PIPING & PRESSURE TEST					
ROUGH HEATING & VENTILATION					
ROUGH FRAMING					
INSULATION					
WALLBOARD					
Other:					
Other:					
Other:					

BUILDING INSPECTION RECORD

Land Use Department ° Building Safety & Inspection Division
Courthouse Annex ° 2045 13th St ° 13th & Spruce Streets
P.O. Box 471 ° Boulder, Colorado 80306-0471

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Permit Number
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DO NOT POUR CONCRETE OR COVER ANY WORK UNTIL APPROVED BY INSPECTOR

FINAL BUILDING INSPECTIONS

INSPECTION	DATE APPROVED	INSPECTOR	COMMENTS	INSPECTOR	DATE
GRADING & DRAINAGE					
FINAL FRAMING					
ELECTRICAL FIXTURES			COVER	JK	12/24/07
			SEE NOTICE		
PLUMBING FIXTURES					
FINAL HEATING & VENTILATION					
Other:					
Other:					
Other:					
BUILDING INSPECTIONS COMPLETED					
ALL REQUIREMENTS COMPLETED					

Use or occupancy is not permitted until all building inspections have been completed and approved and all other required inspections and approvals have been obtained. See also the "Requirements & Remarks" page, printed on the opposite side of the "Building Permit" page, for details on what other requirements may exist.

COVER JK 12/24/07

METER RELEASES

INSPECTION	DATE APPROVED	INSPECTOR	AGENCY	CONTACT	RELEASE DATE	BY
TEMPORARY CONSTRUCTION ELECTRIC METER RELEASE						
FINAL ELECTRIC METER RELEASE	12/24/07	JK	POUDRE		12/26/07	JK
FINAL GAS METER RELEASE						

Building division staff typically informs utility companies of released meters on the business day following the meter release inspection.

All requests for inspection(s) must be received prior to 3:30 p.m., business days, in order to be scheduled for the following business day. Monday inspections must be scheduled by 3:30 p.m. Friday. Please provide the following information: *PERMIT NUMBER, PROJECT ADDRESS, OWNER'S NAME, TYPE OF INSPECTION and DATE OF INSPECTION...* CALL: *(303) 441-3925, press 1, then 2 to schedule an inspection.*